One Grain Too Many: Soil Salinization, Crop Yields, and Rural Adjustment in Brazil

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Abstract

This thesis examines the relationship between soil salinization and Brazilian agriculture from 1985 to 2018. It combines municipal crop data, annual salinity maps, and agroclimatic controls. The Spatial First-Differences strategy compares strictly contiguous municipalities within the same state and watershed. A 1 dS/m increase in excess salinity is associated with approximately 7.8% lower corn yield and 19% lower corn output in the preferred specification. Estimates for the other crops are less precise. Across all crops, a 1 dS/m difference in average salinity is also associated with an annual change in the cultivated share that is 0.065 percentage points lower. Yields are less sensitive in municipalities with higher historical salinity for corn, rice, cassava, and sugarcane. This pattern is consistent with gradual adaptation, although the data do not identify the underlying practice. To study adaptation in rural areas, this dissertation examines changes in crop composition, the area planted per crop, and total cultivated area, as well as internal migration patterns and producer price responses. These findings document agricultural losses and several margins of rural adjustment, but the aggregate data do not track individual farms, plots, or households and do not by themselves establish causal effects.

Keywords: soil salinity, crop yields, agricultural adaptation, agricultural land use, internal migration, spatial first differences, Brazil.

JEL Classification: O13, Q12, Q15, Q24, R23.

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Contents

1 Introduction

Salinization is the excessive accumulation of salts in the root zone of soils, a process that can be linked to local geology, saltwater intrusion, or irrigation and water use practices that increase evaporation and salt concentration. This phenomenon is now one of the main threats to agriculture in many parts of the world. In Brazil, Hassani et al. (2020) estimate that between 1980 and 2018, the country experienced one of the sharpest annual increases in salinized land, with more than 5,600 km^2 added each yearequivalent to approximately 0.07% of the country's total area annually. According to the FAO, 25 to 30% of irrigated land is already affected, particularly in regions dependent on irrigation. This observation is consistent with several assessments by Embrapa in irrigated areas (Cordeiro, 2001; Cordeiro et al., 1988; Cordeiro and Mansefú, 2001).

The main objective of this thesis is to understand how agricultural production and rural populations are affected and adapt to this phenomenon of gradual soil degradation. In order to clearly understand the mechanisms that lead to farmers' actual behaviors, the analysis proceeds in two successive stages. First, I estimate the relationship between soil salinization and municipal crop yields. Second, after quantifying the overall loss of productivity due to salinity, I examine the adjustment margins documented in the data, notably internal migration and changes in agricultural land use. Abandoning land is not necessarily a beneficial adaptation. It may be a strategy to limit losses, but it can also be a direct symptom of economic damage.

The analysis is based on several complementary data sources concerning salinity. The global soil salinity maps produced by Hassani et al. (2020) provide a high spatial resolution annual series covering 1985-2018. I use MapBiomas for land use data, to target salinity data on agricultural areas. These sources are complemented by annual municipal agricultural production data from the PAM survey (*Produção Agrícola Municipal*, since 1974), census microdata (IPUMS and original IBGE database) to track household adaptation behaviors, and high-resolution climate data necessary for the construction of agroclimatic controls.

The main empirical challenge concerns the endogeneity of soil salinity. Salinization is a cumulative stock that reflects natural characteristics, past agricultural practices, and contemporary productive decisions that are imperfectly observed. Standard panel models, or specifications based only on lagged salinity as in Chen and Mueller (2018), are therefore not sufficient to rule out omitted-variable bias.

To address this issue, I use a Spatial First-Difference (SFD) strategy (Druckenmiller and Hsiang, 2018). By exploiting variation between neighboring municipalities, this methodology aims to absorb unobserved and slowly varying structural factors, such as deep soil texture, baseline topography, institutional rigidities, or persistent local practices.

By documenting the impact of changing salinization and adaptation responses, this thesis aims to clarify the strategies implemented by rural populations in the face of growing environmental constraints and to contribute to discussions on agricultural and socioeconomic adaptation to slow environmental phenomena in a major emerging agricultural country.

This study lies at the intersection of three distinct yet complementary strands of the economic literature on salinization. The first strand of literature focuses on the economic impacts of soil degradation and environmental quality on agricultural productivity. Russ et al. (2020) provide a global quantification of the impact of water salinity on agricultural productivity, while Kan (2008) documents the relationship between yield and salinity for tomatoes grown for processing in California. This paper extends this literature to the Brazilian context and to a panel data framework.

The second strand focuses on climate shocks and agricultural adaptation. Numerous studies document the impact of temperature- and precipitation-related shocks on crop yields (Carter et al., 2018), whereas comparatively few economic studies examine soil salinization as an environmental constraint.

The third strand focuses specifically on salinity and migration. Chen and Mueller (2018) examine the impact of coastal salinization in Bangladesh and show effects on agricultural yields and internal mobility. Haque (2025) strengthens identification by exploiting a hydrogeological discontinuity in the Bengal Delta. The Brazilian context requires a different strategy because the exposure does not present the same sharp spatial discontinuity.

Overall, this thesis makes three contributions to the existing literature. First, it provides estimates of the impact of soil salinization on agricultural yields in Brazil. Second, it uses the Spatial First-Difference strategy to estimate the impact of soil salinization on yields while addressing the endogeneity of this phenomenon. Third, it documents the range of productive adjustments observed in response to salinity.

2 Data and Method

2.1 Data

2.1.1 Soil salinity data

The primary measure of soil salinity comes from the global maps developed by Hassani et al. (2020). These data provide annual estimates of soil electrical conductivity (ECe, expressed in dS/m) at a spatial resolution of 1 km $\times$ 1 km, covering the period 1985-2018 in the data used. The model combines remote sensing data such as vegetation indices, Landsat data, and soil and meteorological variables with a Random Forest model trained on globally collected salinity samples. [1]

For the analysis, I aggregate these salinity data at the municipal level using a static

[1] The available salinity files do not allow the main regressions to be extended beyond 2018, even though PAM data are available after that date.

agricultural mask constructed from MapBiomas land-use data. More specifically, I identify and select all 1 km $\times$ 1 km pixels that were classified as agricultural at least once during the entire study period. I then calculate the annual average electrical conductivity within each municipality by strictly restricting the sample to these historically cultivated areas. This fixed-mask approach is crucial to the identification strategy: it limits the risk of endogeneity associated with annual variations in the agricultural boundary, while ensuring that the measure captures exposure over usable agricultural potential rather than the average salinity of non-arable areas.

These data warrant special attention due to the risk of measurement error. The density of soil samples used to train the model is significantly lower in South America than in Europe or North America, which may reduce the accuracy of predictions for Brazil. This measurement error, the direction of which is uncertain, may bias the estimates. Furthermore, since the maps are generated by machine learning, the predictions may smooth out the spatial and annual distribution of salinity and underestimate certain local extremes. Aggregation at the municipal and annual levels may accentuate this smoothing. In this context, the coefficients associated with salinity may be underestimated, particularly when exposure exceeds agronomic thresholds.

Based on these electrical conductivity measures, I construct the main variable of interest, the excess salinity, which is based on the agronomic tolerance thresholds defined by the FAO, which adopts the method described in (Maas and Hoffman, 1977). Rather than using the a raw mean of soil salinity, which does not affect plants in a linear manner, I model the fact that salinity stress only triggers yield losses once a certain biological threshold is crossed. Following agronomic literature, for each crop c, I identify a specific threshold τ_c below which crop development remains unaffected. The variable $Excess\_Salinity_{mct}$ is therefore defined as:

$$Excess\_Salinity_{mct} = \max(EC_{mt} - \tau_c, 0)$$

where EC_{mt} is the aggregated annual electrical conductivity within the agricultural mask of municipality m during year t. I set τ_c to 1.0 dS/m for beans, 1.5 dS/m for cassava, 1.7 dS/m for corn, 3.0 dS/m for rice, 1.7 dS/m for sugarcane, and 5.0 dS/m for soy. This transformation allows us to interpret the coefficient as the marginal impact of each unit of dS/m of excess salt relative to the plant's natural tolerance.

Table 1 documents the distribution of excess salinity in the main regression samples. In particular, it shows that soybeans are a special case: only 7.3% of these crop-municipality-year pair exceed their FAO threshold, and only 9.8% exhibit a nonzero spatial difference in excess salinity.

Table 1: Salinity Exposure Diagnostics in the Main SFD Yield Samples

Crop	Observations	Municipalities	Pairs	Mean ECe (dS/m)	Above FAO
Corn	114,025	4,435	4,070	0.614	55.1%
Rice	66,072	3,660	3,280	0.641	49.0%
Cassava	94,296	4,282	3,860	0.622	55.3%
Beans	102,376	4,400	4,027	0.618	56.1%
Soybeans	33,563	2,004	1,774	0.592	7.3%
Sugarcane	57,069	3,536	3,117	0.623	59.9%

Notes: The table summarizes the estimation samples used in the main specifications of the Spatial First-Differences (SFD) method for agricultural yields. An observation corresponds to a pair of municipalities-crop-year included in the main regressions on yields, after checking that the yields are strictly positive and that the covariates required for the SFD specification have no missing values. The number of municipalities corresponds to the number of unique IBGE municipality codes observed at least once in each crop sample. The mean value of ECe is expressed in dS/m. "Above the FAO threshold" refers to the proportion of crop-year pairs where salinity exceeds the FAO crop-specific agronomic threshold.

2.1.2 Municipal Agricultural Production

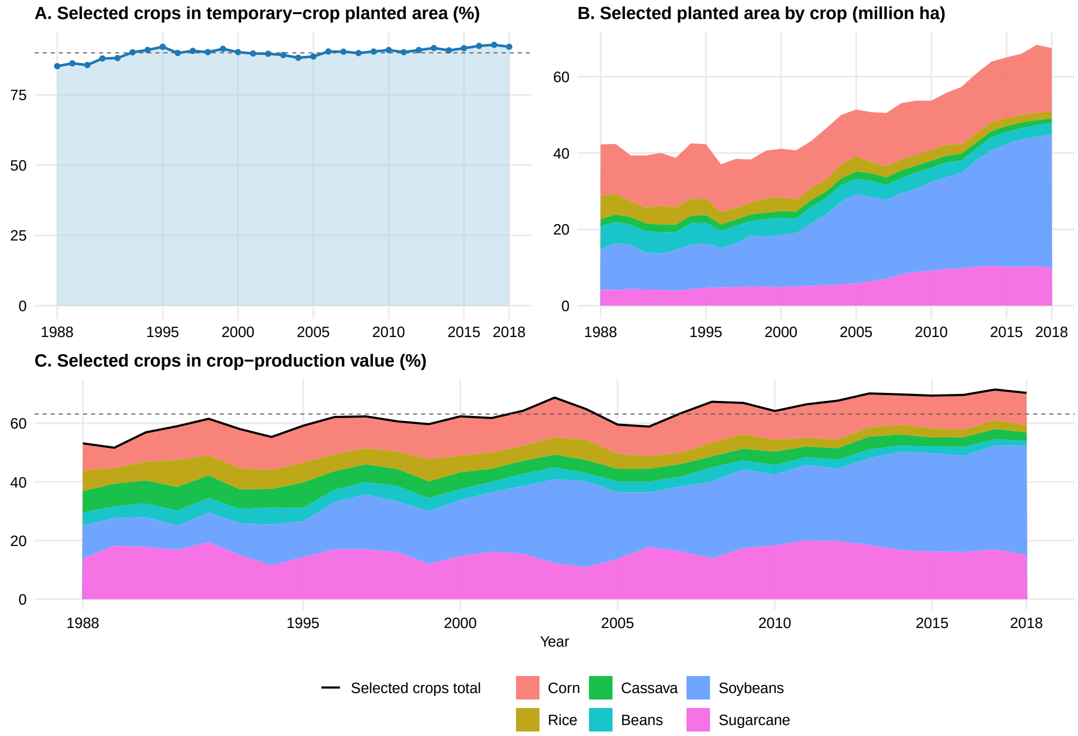
Data. Agricultural production data are drawn from the Municipal Agricultural Production Survey (PAM), published annually by the Brazilian Institute of Geography and Statistics (IBGE) since 1974 (Instituto Brasileiro de Geografia e Estatística, 2024). This survey records, for each municipality and each crop, the planted area, the harvested area, the quantity produced, and the value of production. The estimation sample covers the period 1985-2018 and focuses on six major temporary crops: corn, soybeans, rice, beans, cassava, and sugarcane.

Socioeconomic context. Figure 1 shows that the six crops included in the main analyses constitute a large share of the agricultural sector. In the raw PAM/IBGE data, they account for an average of 90% of the national area planted with temporary crops between 1988 and 2018, and 92.1% in 2018. The figure also shows the distribution of this area by crop. Panel C supplements this information with value data: the six crops accounted for an average of 63.2% of the national production value of temporary and permanent crops reported by PAM/IBGE from 1988 to 2018, and 70.4% in 2018.

These crops refer to different producers and markets. The 2017 agricultural census shows that family farms account for 77% of farms, 23% of their total area and production value, but 67% of agricultural employment. They produce 80% of the value of cassava and 42% of that of beans. Conversely, the IBGE associates large non-family farms with export crops, particularly soybeans and corn (Instituto Brasileiro de Geografia e Estatística, 2017).

Soybeans covered 34.84 million hectares in 2018 and play a central role in agricultural exports (Vieira Filho, 2024). Corn covers 16.55 million hectares. It is also a cash crop, but a large portion of its use is for animal feed, particularly for poultry and pigs (Silva, 2021).

Figure 1: Selected Crops in Brazilian Planted Area and Crop-Production Value

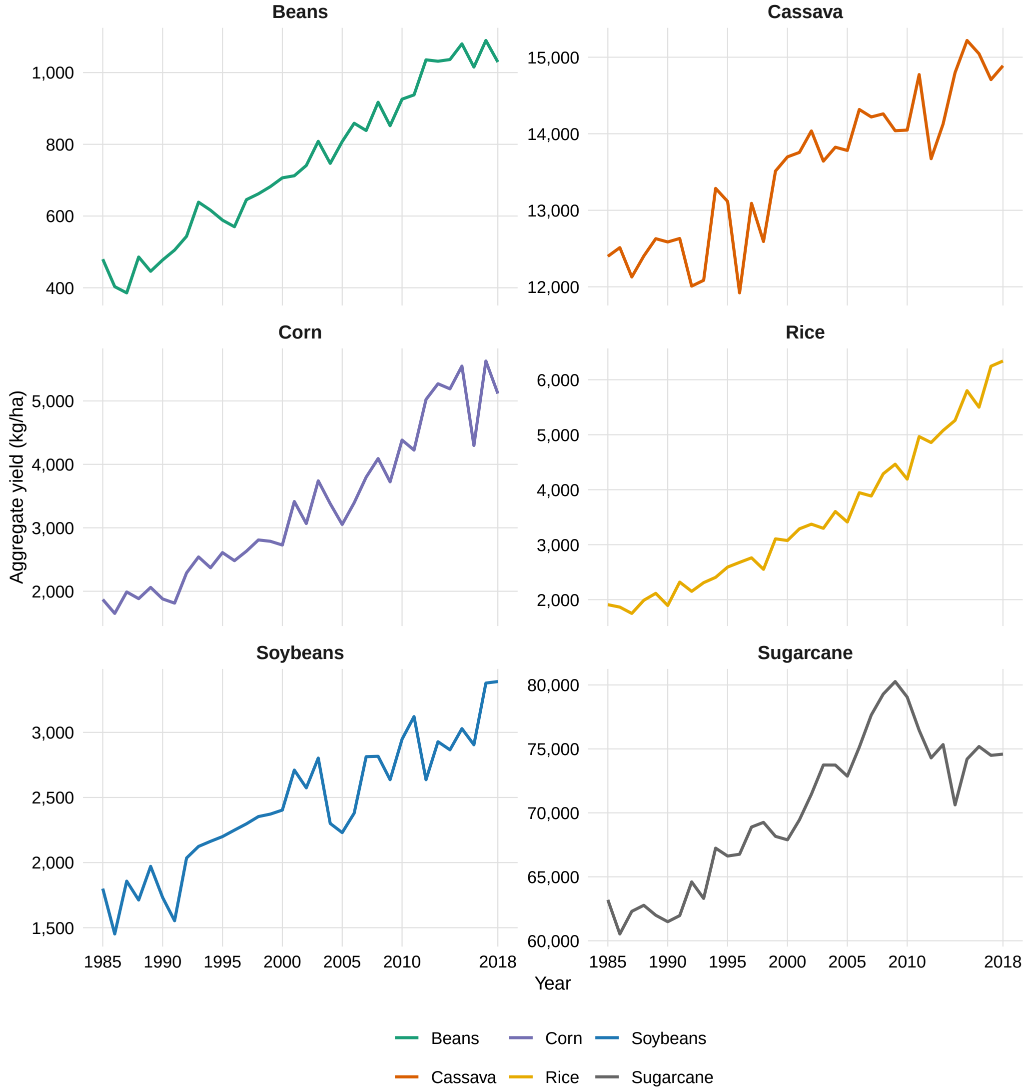


Notes: Panel A reports the selected crops' share of national temporary-crop planted area. Panel B decomposes the selected planted area by crop. Panel C reports the selected crop share of total crop-production value reported by PAM/IBGE.

A decline in yield or production for these two crops can therefore have an impact on national and international supply chains, extending beyond the producing municipalities alone.

Beans and cassava occupy less land, 2.95 and 1.23 million hectares, respectively, in 2018, but they play a direct role in food production and family farming (Wander and Silva, 2025; Rocha et al., 2017; Empresa Brasileira de Pesquisa Agropecuária, 2025c). Sugarcane, covering 10.05 million hectares, is primarily used in the sugar and ethanol industries (Empresa Brasileira de Pesquisa Agropecuária, 2025b). Finally, rice covers 1.88 million hectares and combines upland and irrigated systems, which are heavily concentrated in the South (Empresa Brasileira de Pesquisa Agropecuária, 2025a). These differences in markets, scale, and production systems are taken into account when interpreting the results.

Figure 2: Changes in Yields, 1985-2018



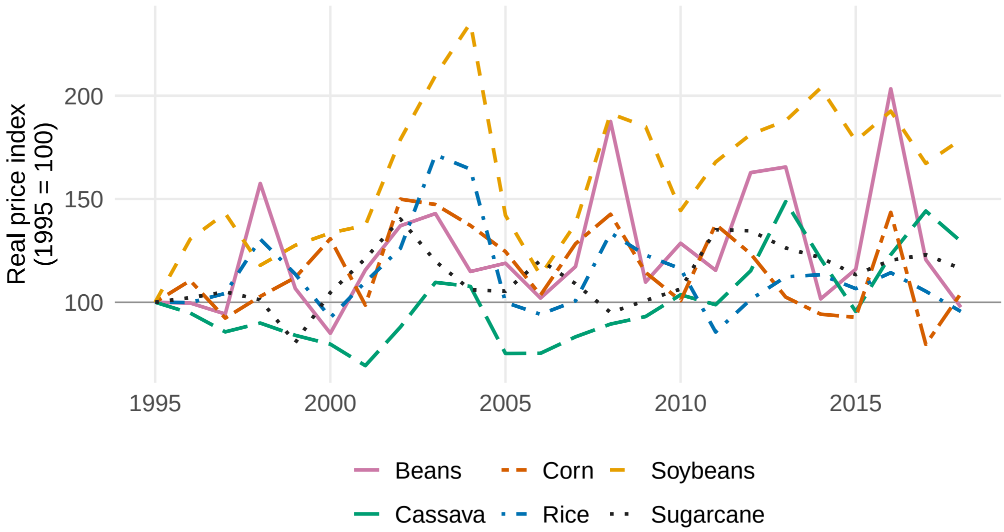
Notes : For each crop and each year, the yield is equal to the sum of the quantities produced, converted from metric tons to kilograms, divided by the sum of the harvested areas. The calculation uses the municipalities included in the PAM panel between 1985 and 2018. Source: IBGE, PAM.

Changes in Yields Yields increased significantly for all six crops between 1985 and 2018. The increase was particularly strong for rice and corn, whose yields in 2018 were approximately 232% and 174% higher, respectively, than in 1985. It was also significant for beans and soybeans, with increases of approximately 115% and 88%. The trends for cassava and sugarcane are weaker, at around 20% and 18%, respectively, and show greater fluctuations around a slow rate of increase. These changes may reflect technological progress, shifts in production locations, and changes in the composition of agricultural systems. However, these yields are increasingly threatened by the effects of climate change, which is exacerbating salinization in particular.

Crop Price Index. PAM/IBGE data are also used to calculate the average producer price for each municipality, crop, and year. The IBGE calculates the value of production based on the quantity produced and the average price paid to producers. The ratio of value to quantity therefore reflects this average price. For the descriptive figure, national prices are deflated using the IPCA (*Índice Nacional de Preços ao Consumidor Amplo*) and then standardized to 1995 = 100. Years prior to 1995 are not shown due to currency changes.

Figure 3 shows different trends across crops. Between 1995 and 2018, the real index reached approximately 179 for soybeans, 129 for cassava, and 116 for sugarcane. Corn has returned to a level close to that of 1995, with an index of 104, while the indices for beans and rice are close to 100, at 98 and 96, respectively. However, the time series remain volatile, particularly for beans, corn, and rice.

Figure 3: National Real Producer-Price Indices by Crop



Notes: For each crop and year, the national average producer price is computed as total PAM production value divided by total production quantity. Prices are deflated by the annual mean of the official IPCA and normalized to 1995 = 100. Years before 1995 are not plotted because of nominal currency changes before the Real. Source: IBGE, PAM and SIDRA Table 1737.

Table 2: PAM Descriptive Statistics by Crop in the Main Yield Regression Sample

Crop	Variable	Mean	SD	Min	P25	Median	P75	Max	N
BEANS	Yield (kg/ha)	719.71	492.57	3.00	390.00	600.00	933.00	7,260.00	102,376
	Production quantity	616.00	2,139.16	1.00	43.00	140.00	432.00	97,410.00	102,376
	Production value (nominal)	38,233.92	675,446.87	0.00	62.00	235.00	1,101.00	95,827,200.00	102,376
	Harvested area (ha)	894.28	2,108.62	1.00	80.00	260.00	800.00	60,368.00	102,376
	Planted area (ha)	929.64	2,189.98	1.00	80.00	270.00	820.00	54,302.00	95,334
CASSAVA	Yield (kg/ha)	14,011.38	5,711.71	80.00	10,000.00	13,000.00	17,500.00	67,222.00	94,296
	Production quantity	4,268.38	10,490.49	1.00	300.00	1,050.00	3,615.00	270,509.00	94,296
	Production value (nominal)	48,752.85	675,783.57	0.00	80.00	319.00	1,467.25	67,632,400.00	94,296
	Harvested area (ha)	310.51	697.73	1.00	25.00	80.00	280.00	17,208.00	94,296
	Planted area (ha)	310.22	695.83	1.00	25.00	80.00	280.00	16,764.00	88,489
CORN	Yield (kg/ha)	2,597.18	2,015.94	1.00	800.00	2,365.00	3,713.00	12,709.00	114,025
	Production quantity	8,227.49	25,438.81	1.00	300.00	1,500.00	6,600.00	1,218,000.00	114,025
	Production value (nominal)	84,525.98	1,270,770.19	0.00	123.00	660.00	3,668.00	183,363,915.00	114,025
	Harvested area (ha)	2,371.51	5,231.64	1.00	270.00	880.00	2,500.00	250,000.00	114,025
	Planted area (ha)	2,430.40	5,381.20	1.00	300.00	900.00	2,500.00	250,000.00	106,844
RICE	Yield (kg/ha)	2,008.43	1,452.14	6.00	1,200.00	1,600.00	2,217.00	11,502.00	66,072
	Production quantity	1,785.26	9,143.58	1.00	43.00	182.00	720.00	278,598.00	66,072
	Production value (nominal)	42,203.10	986,423.50	0.00	20.00	108.00	936.00	114,191,160.00	66,072
	Harvested area (ha)	565.08	1,724.86	1.00	30.00	120.00	400.00	44,217.00	66,072
	Planted area (ha)	545.98	1,724.37	1.00	30.00	101.00	400.00	44,217.00	60,436
SOYBEANS	Yield (kg/ha)	2,420.98	733.24	30.00	1,920.00	2,400.00	3,000.00	12,000.00	33,563
	Production quantity	24,072.13	49,706.01	1.00	1,280.00	7,296.00	26,775.00	1,083,060.00	33,563
	Production value (nominal)	201,799.14	2,435,443.52	0.00	870.00	5,097.00	22,917.00	133,993,335.00	33,563
	Harvested area (ha)	9,287.98	17,313.55	1.00	600.00	3,300.00	11,000.00	280,000.00	33,563
	Planted area (ha)	9,373.65	17,468.45	1.00	600.00	3,396.00	11,000.00	280,000.00	32,023
SUGARCANE	Yield (kg/ha)	47,238.09	20,933.28	1,000.00	30,000.00	43,200.00	60,000.00	175,000.00	57,069
	Production quantity	90,392.43	324,447.96	1.00	800.00	2,925.00	18,000.00	10,260,000.00	57,069
	Production value (nominal)	213,465.97	4,083,068.48	0.00	48.00	237.00	2,321.00	447,480,000.00	57,069
	Harvested area (ha)	1,309.24	4,589.81	1.00	20.00	75.00	360.00	137,500.00	57,069
	Planted area (ha)	1,298.03	4,542.66	1.00	20.00	75.00	350.00	131,000.00	52,972

Notes: An observation is an east-side municipality–crop–year cell attached to an east-minus-west pair that enters the main log-yield SFD regression sample over 1985–2018. The sample requires strictly positive yield and non-missing salinity, GDD, KDD, soil moisture, soil/topographic controls, pair coordinates and year fixed effects. Values shown are east-side PAM variables; the corresponding west-side municipality is also observed because the SFD difference is defined. Production value is nominal in the monetary unit reported by PAM/IBGE.

2.1.3 Weather Data

The meteorological data come from the BR-DWGD dataset developed by Xavier et al. (2022), which provides daily data interpolated on a 0.1 degree grid covering the entire territory of Brazil since 1961.[2] I aggregate these data at the scale of each municipality-crop-year growing season. [3]

To accurately capture the nonlinear physiological responses of crops to daily temperature fluctuations, I use the concepts of Growing Degree Days (GDD) and Killing Degree Days (KDD), following Schlenker and Roberts (2009). Crop development depends on thermal thresholds: a base threshold (T_{base}) below which vegetative growth ceases, and an upper cutoff threshold (T_{kill}) above which heat stress becomes damaging.

To model the thermal profile using only daily extreme data (T_{max} and T_{min}), I apply a continuous sinusoidal specification to each pixel in the grid, adapting the method from Snyder (1985). For each crop, the optimal physiological limits and severe heat stress thresholds are calibrated based on the agronomic synthesis in Paredes et al. (2025).

Specifically, daily GDD accumulation represents the thermal time spent in the interval favorable to crop growth, while KDD accumulates exposure to severe heat stress. These daily exposures are first calculated at the pixel level and then aggregated to the relevant growing season for each municipality-crop pair.

2.1.4 Migration Data

The analysis uses microdata from Brazilian censuses harmonized by IPUMS International (IPUMS International, 2026c,d). The comparable geographic unit is a municipal geography harmonized by IBGE. It does not always correspond to a single municipality, as IBGE combines some small municipalities with neighboring municipalities to maintain comparable boundaries over time.

An origin location is defined as rural when the weighted proportion of its population living in a rural area is at least 50%. A regression observation is an origin-destination pair observed during an five-year migration period. The time since arriving at the destination allows migrants to be divided into two categories per census.

For a period t, the bilateral flow is simply defined as:

$$m_{t,o,d} = \sum_{i \in \mathcal{C}(t)} w_i \, \mathbf{1}\{O_i = o,\ D_i = d,\ R_i \in I_t\}, \tag{1}$$

where $\mathcal{C}(t)$ denotes the census used and w_i the individual's weight. For individual i, O_i is the last municipality of residence, D_i is the current municipality, and R_i is the time

[2] This data source differs from the international data often used because it was developed specifically for Brazil and is directly available at a higher spatial resolution.

[3] The planting and harvesting calendars used here are spatialized but are not directly observed at the municipal level. Municipal variation arises from the intersection of these calendars, climate pixels, and the crops grown in each municipality.

since arrival. Finally, I_t is the time interval associated with the period. Durations of 5 to 9 years correspond to the periods 1981-1985 and 2001-2005. Durations of 0 to 4 years yield the windows 1986-1990 and 2006-2010. This construction measures the most recent reported intermunicipal move. It does not allow for the observation of a previous migration, a return migration, or the exact date of the move within the time window.

The data do not provide information on the economic activity carried out in the last municipality of residence prior to relocation. They indicate only the sector of employment reported at the time of the census, which is at the destination and after displacement (IPUMS International, 2026b,a). The main analysis therefore focuses on all individuals who leave a rural area. It does not assume that they were farmers before their departure.

For each origin, the exposed population is the weighted population aged 15 and older in the reference census: 1980 for the 1991 periods and 2000 for the 2010 periods. The dependent variable remains the weighted bilateral migrant flow, while origin population enters the PPML as an offset. The coefficients can therefore be interpreted as proportional changes in the expected bilateral migration flow within the retained dyads.

Table 3: Construction of Municipal Migration Flows

Migration window	Census	Residence duration	Main regression	Sampled migrants	Weighted migrant flow	Positive corridors	Origins	Destinations
1981-1985	1991	5–9 years	No	278,366	2,446,236	46,813	1,115	2,040
1986-1990	1991	0–4 years	Yes	398,807	3,478,209	60,401	1,115	2,040
2001-2005	2010	5–9 years	Yes	88,990	829,932	21,855	482	1,994
2006-2010	2010	0–4 years	Yes	150,039	1,371,445	30,103	482	2,026

Notes: Flows sum census person weights for people aged 15 or older whose last municipality differs from their municipality at the census. Origins are restricted to harmonized areas whose weighted rural population share was at least 50 percent in the baseline census. The 1981–1985 flow is constructed and reported but excluded from the regression because the salinity maps begin in 1985 and do not provide a predetermined exposure measure. Both duration bins approximate five-year windows because the census reports completed years in the current municipality rather than an exact move date. The 5–9-year windows may be more affected by return migration and attrition than the 0–4-year windows.

The estimation panel is restricted to 37,141 dyads with at least one positive flow during one of the three estimated windows. Within these retained dyads, zero-flow cells are preserved and account for 55,998 of the 111,310 regression observations. Dyads with no positive flow in any window are not included. The model therefore does not represent migration to every possible destination, and this selection limits the external scope of the estimates.

2.2 Empirical Strategy

In this section I discuss how I explore empirically the link between soil salinity and different outcomes. These outcomes can be modelled as reduced form functions of an agricultural outcome y and a set of relevant parameters such as temperature or soil moisture. Assuming that y is a function of local weather and other factors, I can approximate these reduced forms using the following log-linear regression model:

$$\ln y_{mct} = \beta_c S_{mct} + \Phi_c' X_{mct} + \alpha_m + \lambda_t + \varepsilon_{mct} \tag{2}$$

where the unit of observation is municipality m, crop c, and year t, and y is an outcome such as agricultural yields, output or planted area. S_{mct} denotes the soil salinity measure. It takes the form of either average gross salinity or salinity in excess of the agronomic threshold. X_{mct} represents agroclimatic controls and other observed controls. α_m and λ_t are municipality and year fixed effects.[4]

However, estimating equation (2) raises a significant identification problem. Salinization is a persistent and cumulative phenomenon, determined jointly by the natural characteristics of the territory and past agricultural practices. Deep soil texture, drainage conditions, geology, topography, water infrastructure, and even the local history of irrigation can thus simultaneously influence both salinity and agricultural yields. A level-based estimation therefore risks attributing to salinity the effect of unobserved territorial determinants. Municipal fixed effects make it possible to eliminate characteristics that are strictly invariant over time. However, identification then relies exclusively on temporal variations in salinity within a single municipality. Yet these annual variations are relatively small and persistent. This low intra-municipal variation can reduce the accuracy of estimates and amplify the attenuation bias associated with measurement errors in the predicted salinity maps.[5]

To reduce the influence of unobserved local determinants, I apply the Spatial First Differences (SFD) method proposed by Druckenmiller and Hsiang (2018). This approach compares differences in outcomes and salinity between geographically neighboring municipalities, rather than relying solely on differences between distant municipalities or temporal variations within a single municipality.

Let $p = (i, j)$ be a pair of two neighboring municipalities. For any variable V, the spatial difference operator is defined as:

$$\Delta_s V_{pt} = V_{it} - V_{jt}. \tag{3}$$

For each crop c, the estimated equation is then:

$$\Delta_s \ln y_{pct} = \beta_c \Delta_s S_{pct} + \Phi_c' \Delta_s X_{pct} + \lambda_{ct} + u_{pct}, \tag{4}$$

The main pairs are constructed along a west-east orientation. For each municipality, the algorithm identifies a neighboring municipality immediately to the west, provided that both units belong to the same state, the same hydrological basin, and the same

[4] A municipality is the smallest administrative jurisdiction in Brazil. The sample includes 5,425 municipalities out of a total of 5,570. The remaining municipalities are excluded before estimation because they are not jointly represented in the intermediate agricultural and spatial panel. Most exclusions reflect the absence of a usable salinity measure under the broad agricultural mask. The remainder arise from unavailable agricultural records.

[5] An instrumental variable approach was considered, but it is very difficult to find a variable that strictly satisfies the exclusion restriction.

latitude[6]. The restrictions regarding the state and the hydrological basin are intended to compare municipalities subject to institutional and hydrological frameworks that are as similar as possible. The difference is consistently oriented within the sample: for each pair, the value observed in the municipality located to the east is compared to that of the municipality located to the west.[7] The state $\times$ hydrological basin restriction is conservative. The state restriction limits comparisons between municipalities subject to different administrative, fiscal, and agricultural policy frameworks. The hydrological basin restriction limits comparisons between municipalities whose conditions regarding drainage, salt accumulation, and hydrological transport are less comparable. The model therefore does not merely compare municipalities that are geographically close. It compares strictly contiguous municipalities within the same administrative and hydrological framework. In the PAM panel with zeros, 4,306 west-east pairs are selected. Of the 5,425 available municipalities, 4,676 appear at least once in a final pair, 749 never appear in a final pair, and 1,119 do not qualify as an eastern municipality with a western neighbor.[8]

The causal interpretation of β_c relies on a conditional local orthogonality assumption. Formally, I assume that:

$$\mathbb{E}\left[\Delta_s S_{pct}\, u_{pct} \,|\, \Delta_s X_{pct}, \lambda_{ct}\right] = 0. \tag{5}$$

This condition does not require that two neighboring municipalities be identical. It does, however, require that, after accounting for observed differences in climate, topography, and geological characteristics, local variations in salinity not be systematically associated with variations in unobserved determinants of yields.

The plausibility of this hypothesis rests on the idea that many structural determinants of agricultural productivity vary relatively continuously across space. When two neighboring municipalities are compared, differences in geology, general climatic conditions, market access, prices, or the institutional environment should be smaller than in a comparison between distant municipalities. This is supported by Table 8: within-pair variation in observed covariates falls to only 16.9–28% of the benchmark dispersion, while salinity retains 46.1% of its benchmark variation (48.1% for excess salinity). Thus, the SFD substantially improves comparability between municipalities while preserving meaningful local variation in salinity for identification.

However, this strategy does not eliminate all potential sources of endogeneity. In particular, the identification hypothesis could be violated if local differences in irrigation, drainage, input use, farm composition, or agricultural policies simultaneously influence both salinity and yields. The estimates can therefore be given a causal interpretation

[6] Latitude here refers to a band of 0.43 degrees used to avoid comparing neighboring municipalities that are located on north-south gradients that differ too greatly.

[7] To assess the sensitivity of the results to the choice of spatial orientation, an alternative construction of the pairs along a north-south axis is used as a robustness test.

[8] These numbers reflect the restrictions regarding strict contiguity, state boundaries, hydrological basins, and latitude.

only under the assumption that these unobserved determinants are not correlated with the residual differences in salinity between paired municipalities.

I do not include contemporaneous municipal prices as controls in the agricultural-outcome specifications, since they may themselves respond to salinity and agricultural output. The SFD design nevertheless removes price components common to the two neighboring municipalities in a pair, while year fixed effects in the crop-specific regressions absorb national time-varying shocks common to each crop, including national crop-price movements. Consistent with the local nature of these comparisons, the average spatial difference in log prices is very close to zero for every crop in the estimation samples, as shown in Table 8. This evidence does not rule out differential local price shocks, which may still remain a source of omitted-variable bias. I therefore also examine municipal prices directly as an outcome.

According to Druckenmiller and Hsiang (2018), Spatial First Differences reduce the unobserved heterogeneity shared among immediate neighbors, but the inference must still allow for spatial autocorrelation of the differentiated residuals. I therefore use the spatial estimator from Conley (1999), with a principal radius of 200 km. This radius remains consistent with a local agronomic interpretation: it does not define exposure to salinity, which always comes from contiguous municipalities, but it allows for residual dependence related to climatic, hydrological, or measurement shocks that are spatially correlated on a limited regional scale. Tables 14 and 16 report robustness checks for the main yield regressions using cutoffs of 50, 100, 200, 250, and 300 km for the Conley spatial standard errors. The 300 km radius is used as a wide sensitivity check, not as the main inference window, because several Conley variance-covariance matrices require a numerical correction to be positive semi-definite.

Three specifications are primarily used. Specification 1 includes only the salinity variable. Specification 2 is the preferred option. It accounts for the specific weather conditions of each survey. Differences in soil and topography are included in Specification 3, but these are fixed and should already be largely accounted for by comparing immediately adjacent municipalities.

In equation (4), I am interested in β_c: the reduced-form estimate of the effect of salinity on agricultural outcomes for crop c. At the core of the analysis is the assumption that soil salinity affects productivity. I first estimate regression (4) using agricultural yields as dependent variable. A main limitation of this approach is that yields are a partial measure of productivity. They relate production to the harvested area, but do not measure total inputs or the planted land that is ultimately not harvested. They should therefore not be interpreted as total factor productivity (Aragón et al., 2021, 2022).

2.2.1 Modeling the relation between soil salinity and yields

Following established agronomic and economic literature (Maas and Hoffman, 1977), I model the relationship between soil salinity and agricultural productivity as a function of the crop's physiological stress induced by salt accumulation in the root zone. In the main specification, S_{mt} is excess salinity above the crop-specific threshold. In the alternative specification, S_{mt} is raw mean salinity.

$$\beta_c S_{mct} = \beta_c \text{Excess\_Salinity}_{mct}$$

This transformation allows the coefficient β_c to be interpreted as the marginal impact of an additional unit of 1 dS/m of salt stress relative to the crop's natural physiological tolerance.

As an alternative specification and robustness check, I also estimate the model using raw mean salinity ($\overline{EC}_{mt}$) instead of excess salinity:

$$\beta_c S_{mct} = \beta_c \overline{EC}_{mt}$$

While mean salinity does not explicitly incorporate biological thresholds, this alternative formulation ensures that the baseline findings are not driven by the functional form imposed by τ_c or by the concentration of observations at zero when salinity levels remain below crop tolerance limits.

The choice between average salinity and excess salinity therefore depends on the dependent variable. When the dependent variable is specific to a crop, such as corn yield or rice production, excess salinity is the primary measure. It corresponds to biological stress above that crop's tolerance threshold. When the dependent variable is not specific to a single crop, such as migration, total planted area, or net abandonment, average salinity is a more consistent measure. In such cases, there is no single tolerance threshold that would apply to all crops and all local agricultural activities.

2.2.2 Historical Exposure and Adjustment of Yields: Model

Are the municipalities most at risk those that are least sensitive to soil salinity? The answer to this question would support the hypothesis that farmers adapt to the process of soil salinization.

Here, I adopt the two-step approach proposed by Butler and Huybers (2013). This approach is also formalized by Auffhammer (2022), which first estimates local response functions based on temporal variations and then links their slopes to long-term climatic conditions. The authors first estimate, for each county, the temporal sensitivity of yield to temperature anomalies. They then relate this sensitivity to the county's historical average climate. The idea is simple: if producers have adapted to a persistent constraint, areas that have historically been more exposed should be less sensitive to an additional variation in that constraint.

In the first step, I estimate a separate regression for each municipality and each crop:

$$\ln Y_{mct} = a_{mc} + \lambda_{mc} t + b_{mc}\Big(Excess^c_{mt} - \overline{Excess}^c_m\Big) + \Gamma'_{mc}\widetilde{W}_{mct} + u_{mct}. \qquad (6)$$

The variable in parentheses is the annual anomaly of crop-specific excess salinity, calculated relative to its mean in the municipality-crop series. The vector $\widetilde{W}_{mct}$ contains the standardized anomalies for GDD, KDD, and soil moisture. The term $\lambda_{mc}t$ accounts for a trend specific to each series. The coefficient b_{mc} therefore measures the temporal sensitivity of yield to an additional 1 dS/m of excess salinity in that municipality. I retain the time series with at least fifteen years of positive yield between 1985 and 2018 and the municipalities with at least five salinity observations between 1985 and 1994. The standard error of $\widehat{b}_{mc}$ is calculated using bootstrap method.

In the second step, I use the logarithmic form of their approach:

$$\widehat{b}_{mc} = \alpha_{0c} + \alpha_{1c} \ln\left(\overline{EC}^H_m\right) + \eta_{mc}, \tag{7}$$

where $\overline{EC}^H_m$ is the municipality's average gross salinity between 1985 and 1994. This measure describes the historical salinity environment common to all crops, while the first step uses the agronomic stress specific to each crop. The regression is weighted by the inverse of the bootstrap variance of $\widehat{b}_{mc}$. A coefficient $\alpha_{1c} > 0$ means that the slope becomes less negative as historical exposure increases. This result would be consistent with adaptation.

2.2.3 Adaptive-response specifications

I distinguish between the effect of salinity on crop productivity and the productive responses that may result from this shock. The yield regressions measure production per hectare of harvested land. They do not allow me to determine whether producers also adjust the scale of production, crop composition, or agricultural land use. That is why I estimate a second set of specifications in which the dependent variables describe production margins rather than yield itself.

First, I use the same crop-specific SFD framework as in equation (4), but I replace yield with physical output and harvested area. These outcomes are observed at the end of the growing season. They are therefore estimated based on contemporary excess salinity and the same three sets of control variables used in the yield regressions. The regressions on production and harvested area allow us to verify whether yield losses translate into changes in total production or in the area actually harvested. They should not be interpreted as purely ex ante choices of inputs, since the harvested area may also reflect crop failures or abandonment during the growing season.

Second, I examine the crop composition using the shares of planted area. This measure is closer to a land-allocation decision, since the planted area is observed before harvest. For crop c, municipality m, and year t, I define:

$$Share^P_{mct} = 100 \times \frac{PlantedArea_{mct}}{\sum_k PlantedArea_{mkt}}, \tag{8}$$

where the denominator corresponds to the total planted area of the municipality for all crops observed in the PAM and not just the selected crops.

For the productive margins specific to a crop, I follow exactly the SFD logic of equation (4). I only change the dependent variable. Instead of using the logarithm, which would eliminate zero areas even though it provides information about crops that are disappearing, I preferred to use the inverse hyperbolic sine function, *asinh*, which approximates the logarithm. For the quantity produced and the harvested area, I estimate:

$$\Delta_s \operatorname{asinh}(Y^r_{pct}) = \beta_c \Delta_s S^c_{pct} + \Phi'_c \Delta_s X_{pct} + \lambda_{ct} + u_{pct} \tag{9}$$

where the term S^c_{pct} represents the excess salinity above the FAO threshold for crop c. The asinh function preserves the zeros while producing a curve that closely resembles the *log* function for positive values, but its coefficient is close to a semi-elasticity only for sufficiently large positive values. The zero points require a separate interpretation (Bellemare and Wichman, 2020).

For the proportion of planted area, I retain the same structure but shift the salinity by one year:

$$\Delta_s Share^P_{pct} = \beta_c \Delta_s S^c_{pc,t-1} + \Phi'_c \Delta_s X_{pc,t-1} + \lambda_{ct} + u_{pct}. \tag{10}$$

This difference is important. The quantity produced and the harvested area are observed after the growing season. They may therefore include losses that occurred during the season. The planted area is closer to a decision made before harvest. The three specifications remain the same as for yields: fixed effects for the year, followed by GDD, KDD, and soil moisture, and finally controls for soil and topography.

2.2.4 Total Cultivated Area

This analysis examines whether salinity is associated with a decrease in the area devoted to crops. It does not focus on any particular crop. It includes all annual and perennial crops and excludes pastures and other non-crop land uses. This question differs from the one regarding the areas reported in the annual survey by crop. A decline in corn production may be offset by an increase in soybean production, without reducing the total amount of cultivated land. Furthermore, adding up the PAM areas may result in a plot used for multiple crops in the same year being counted twice. I therefore use the physical area classified as cropland by MapBiomas (Souza et al., 2020).

Let $A^{MB}_{m,t}$ be the cultivated area classified by MapBiomas in municipality m, and G_m its geographic area. I first define the municipal cropland share as $C_{m,t} = 100 \times A^{MB}_{m,t}/G_m$, and its annual change as $\Delta C_{m,t+1} = C_{m,t+1} - C_{m,t}$, measured in percentage points. The regression outcome is the east-minus-west difference in this annual percentage-point change. I estimate the following specification:

$$\Delta_s \Delta C_{p,t+1} = \sum_{h=0}^{H} \beta_h^L \Delta_s \overline{EC}_{p,t-h} + \Phi_L' \Delta_s W_{p,t} + \lambda_t + u_{p,t+1}. \tag{11}$$

The annual specification sets $H = 0$. The distributed-lag model sets $H = 5$ and includes all six salinity values in the same regression. The term $\overline{EC}_{p,t-h}$ is the average salinity, measured in dS/m. I use a common salinity value because the outcome encompasses multiple crops and does not have a single agronomic threshold. The vector $W_{p,t}$ contains GDD, KDD, and soil moisture, calculated as the simple average across the six cropping calendars. Specification 1 contains only the fixed effects for year λ_t. Specification 2, which is the preferred model, adds these meteorological controls. Specification 3 adds elevation, slope, and clay content.

The coefficient β_h^L measures, in percentage points, the difference in annual variation in the proportion of cultivated land associated with an additional 1 dS/m of salinity at lag h. When $H = 5$, each coefficient is conditional on the other five lags. Their sum describes a difference in salinity maintained over six years. It does not measure a cumulative loss of land over six years.

The SFD compares neighboring municipalities, and fixed-year effects absorb national shocks. The inference uses Conley standard errors with a primary radius of 200 km, followed by radii of 50, 100, 250, and 300 km.

2.2.5 Crop-Specific Changes Between Campaigns

This analysis examines whether the planted area or the aggregate yield of a crop in a municipality changes between growing season t and growing season $t + 1$ following an increase in excess salinity in t. In the annual PAM data, $t + 1$ is the subsequent growing season for the same municipality-crop cell.

The planted area must be strictly positive at t, and the quantity produced must be strictly positive at $t - 1$. I do not require positive production at t, since this may already reflect the salinity measured at t. A positive value at $t - 1$ indicates that the crop was recorded prior to the growing season, without directly selecting based on current production. Zero yields at t and $t + 1$, as well as zero areas at $t + 1$, are retained. It only identifies continuity of the crop within the municipality prior to the current survey.

The main dependent variable is the change in asinh between the two surveys. For $Y \in \{A, Q\}$, I estimate separately for each crop:

$$\Delta_s[\text{asinh}(Y_{pc,t+1}) - \text{asinh}(Y_{pc,t})] = \beta_c \Delta_s Excess_{pc,t}^c + \Gamma_c' \Delta_s W_{pc,t} + \lambda_t + \varepsilon_{pc,t+1}. \tag{12}$$

When Y_t and Y_{t+1} are both positive and sufficiently large, the difference of asinh is close to $\log(Y_{t+1}/Y_t)$. The coefficient can then be interpreted approximately as a proportional differential change. This interpretation does not apply when the value crosses zero. If $Y_t > 0$ and $Y_{t+1} = 0$, the change is exactly $-\operatorname{asinh}(Y_t)$. If production is zero at t and then becomes positive again at $t+1$, the change is $\operatorname{asinh}(Y_{t+1})$. These observations are therefore neither omitted nor replaced by a constant.

The estimand thus combines the adjustments for positive values and the zero crossings recorded at the municipal level. It does not represent a pure intensive margin. Nor does it directly measure the abandonment of a farm, since a zero value merely indicates that no area or production for that crop is recorded in the municipality in the following growing season. Conversely, a positive value on both dates does not prove that the same farmers continue to grow the crop: entries and exits may offset each other in the municipal aggregate.

Specification 1 contains only fixed-year effects. Specification 2 is the preferred specification; it adds the spatial differences in their variations between t and $t+1$. This specification places the weather controls on the same interval as the variation in production. Specification 3 adds differences in elevation, slope, and clay content. All three specifications use exactly the same crop-specific sample.

To interpret the planted area, I then distinguish between two categories: the area in use and the exit from an activity following an increase in salinity. The first measure estimates the logarithmic change in area among pairs where municipalities maintain a positive aggregate area at $t+1$. This restriction is determined after data processing. The second measure indicates whether the recorded municipal area becomes zero at $t+1$ and remains so at $t+2$. This represents a persistent absence of cropland in the municipal data.

I also repeat the estimation among the “municipality-crop” cells whose aggregate area and production are positive in $t-1$ and t. This restriction reduces the impact of temporary crop appearances and annual crop rotations, which can create misleading dynamics in the aggregated data. However, it does not transform the annual PAM survey into a producer panel. A placebo model uses the variation in area between $t-1$ and t.

2.2.6 Crop Price Index

This analysis examines whether salinity is associated with a change in the price received by producers. A positive finding could offset part of the physical yield loss. The average producer price is calculated, for each municipality, crop, and year, as the PAM value divided by the quantity produced. This is an average price. It may also reflect quality, crop varieties, and local markets.

For each crop c, I estimate:

$$\Delta_s \log P_{pct} = \beta_c^P \Delta_s Excess_{pct}^c + \Phi_c' \Delta_s W_{pct} + \alpha_p + \lambda_{s(p)t} + \varepsilon_{pct}, \tag{13}$$

where P_{pct} is the average producer price, $Excess_{pct}^c$ is the salinity above the crop's threshold, and W_{pct} represents weather controls. The coefficient β_c^P is a semi-elasticity:

it measures the relative change in price associated with 1 dS/m of additional excess salinity between the two sides of the pair.

The first specification includes fixed year effects and all controls. The second adds fixed pair effects α_p and retains the meteorological controls. The third specification, which is preferred, replaces the fixed year effects with fixed state-year effects $\lambda_{s(p)t}$. It accounts for permanent price differences between neighbors and market shocks common to a state. The standard errors are computed using the Conley spatial estimator with a 200 km cutoff.

2.2.7 Migration Model

I now move to a possible extensive margin of rural adjustment. The question is whether higher salinity changes bilateral out-migration from a given rural origin. The main result relies on a gravity model estimated directly by PPML. Within the retained origin-destination dyads, this method keeps zero flow observations and accounts for the population exposed at origin (Santos Silva and Tenreyro, 2006; Head and Mayer, 2014).

For each origin o, destination d, and migration window t, I estimate:

$$\mathbb{E}[m_{t,o,d} \mid Z_{t,o,d}] = \exp\Big[\log P_{o,b(t)} + \beta_O^M \widetilde{EC}_{o,t-1} + \beta_D^M \widetilde{EC}_{d,t-1} + C'_{t,o,d}\theta^M + \mu_{od} + \lambda_{s(o)t} + \delta_{s(d)t}\Big], \tag{14}$$

Here, $m_{t,o,d}$ is the weighted flow defined in equation (1). It sums the census weights of individuals aged 15 or older at the census whose most recent reported intermunicipal move links o to d during period t. The population $P_{o,b(t)}$, measured in the baseline census before the window, enters as an offset with its coefficient fixed at one. The dependent variable therefore remains a flow, while $\mathbb{E}[m_{t,o,d} \mid Z_{t,o,d}]/P_{o,b(t)}$ is an expected bilateral out-migration rate.

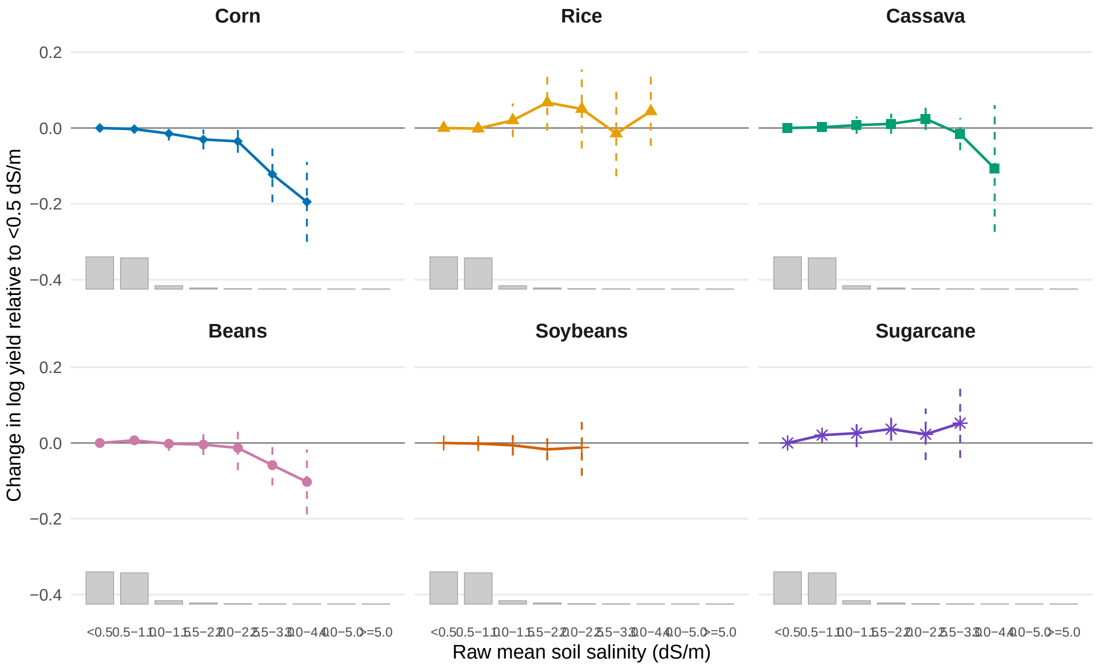
The average salinity levels at origin and destination, $\widetilde{EC}_{o,t-1}$ and $\widetilde{EC}_{d,t-1}$, are measured before the period and standardized separately within each window. Here, $t-1$ denotes the pre-period measurement year rather than the preceding migration period. The vector $C_{t,o,d}$ contains GDD, KDD, and soil moisture, also measured before the period at both origin and destination.

Both specifications include origin-destination dyad fixed effects μ_{od}, origin-state-by-window fixed effects $\lambda_{s(o)t}$, and destination-state-by-period fixed effects $\delta_{s(d)t}$. The first absorb constant bilateral characteristics, such as distance or persistent migration networks. The others absorb shocks common to areas in the same state during each window. The second specification, which is preferred, adds the climate controls at origin and destination. Standard errors are two-way clustered by origin and destination.

The common coefficient β_O^M measures the change in the expected bilateral out-migration rate associated with a one-standard-deviation increase in salinity at origin. Its percentage equivalent is $100[\exp(\beta_O^M) - 1]$. The coefficient β_D^M gives the corresponding association with salinity at destination. This specification compares changes in flows within the same dyad across the three estimated windows. It also conditions on state-level shocks and climate conditions measured before migration. This timing limits the risk that salinity is measured after migration. It does not make unobserved local changes that jointly affect salinity and migration exogenous. Moreover, because the estimation panel retains dyads with at least one positive flow, the model does not describe migration to every possible destination.

3 Results

Figure 4: Crop yield responses across raw mean soil salinity bins

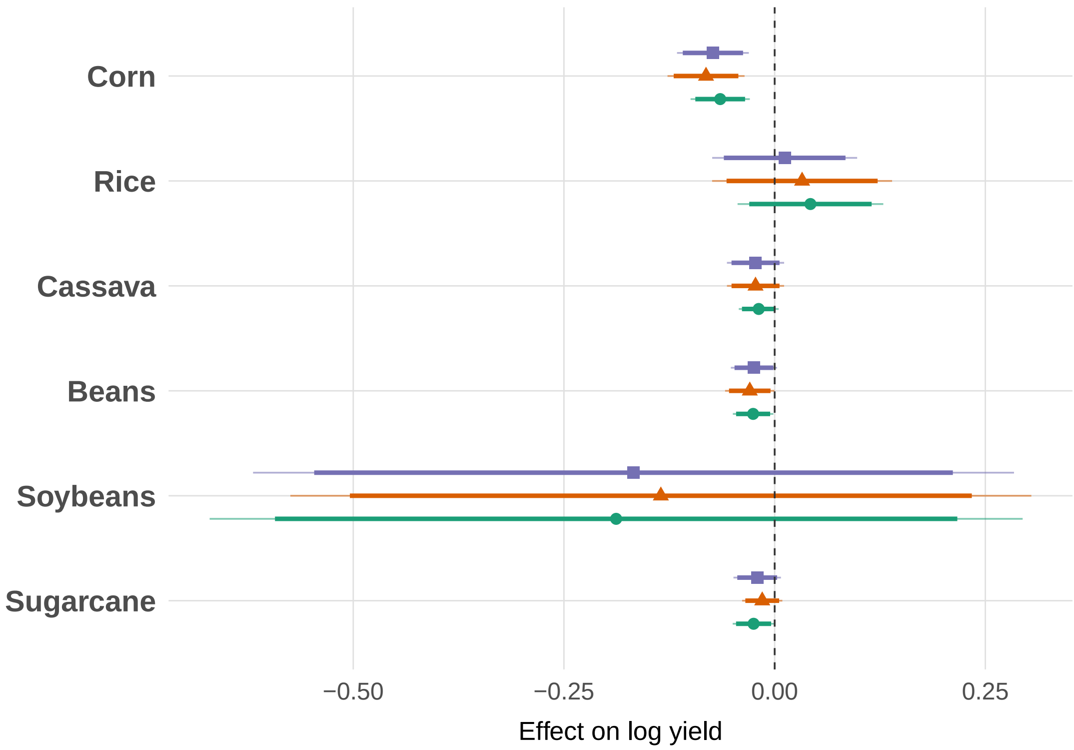


Notes: Points report crop-specific SFD coefficients estimated at the municipality-pair-year level. The horizontal axis shows bins of raw mean soil salinity (dS/m), with salinity below 0.5 dS/m as the omitted reference category. All spatial differences are measured east minus west. Dashed bars show 95% confidence intervals. Models include year fixed effects and controls for GDD, KDD, soil moisture, slope, elevation, and clay content. Standard errors are computed using the Conley spatial estimator with a 200 km cutoff. Grey bars show the mean within-crop salinity distribution in the estimation samples, giving equal weight to each crop.

3.1 Soil salinity and productivity

Figure 4 provides the context for the empirical analysis. It offers preliminary evidence of a nonlinear relationship between soil salinity and agricultural productivity: at moderate levels, salinity has no significant impact on yields, but at higher levels, the effect is negative. This illustrates several things. First, it justifies the use of excess salinity rather

Figure 5: Excess Soil Salinity and Crop Yields



Notes: The figure reports Spatial First-Difference estimates. The dependent variable is the spatial difference in log crop yield, and salinity is measured as crop-specific excess salinity above the FAO agronomic threshold, in dS/m. Green circles show Spec. 1 with year fixed effects only; orange triangles show Spec. 2, which adds GDD, KDD and soil moisture; purple squares show Spec. 3, which additionally adds elevation, slope and clay. Thick lines show 90% confidence intervals and thin lines show 95% confidence intervals based on Conley standard errors with a 200 km cutoff. The five-year zero criterion is applied before estimation; yield remains conditional on strictly positive production and harvested area because log yield is undefined at zero.

than a simple measure of salinity. But it may also be a sign of factors that are less agronomic in nature: as salinity increases, farmers can compensate by using more input, at least up to a certain point.

Figure 5 and Table 12 present the estimates obtained from the excess salinity measure. The first specification includes only annual fixed effects, the second adds agroclimatic controls, and the third extends the model to include soil and topographic characteristics.

The estimates presented in Figure 5 indicate that the effect of salinization on yields varies significantly among crops. The most robust result concerns corn: across all specifications, a local increase in excess salinity is associated with a substantial decrease in its yield. In the most comprehensive specification, a 1 dS/m increase above the agronomic threshold corresponds to an average decrease of approximately 7% in yield. The stability of this estimate after controlling for climate, soil, and topographic factors suggests that the relationship is not primarily explained by observable local differences in climate or soil quality.

This result is consistent with the agronomic mechanisms of salinity. Increased salt concentration reduces the ability of roots to absorb water and can cause ionic imbalances that limit vegetative growth and grain formation. The result does not demonstrate that corn is biologically more sensitive than all other crops. The coefficient can also reflect production systems, irrigation, cropping schedules, or the location of production areas.

Estimates for beans are also negative, but less precisely quantified. They suggest a more moderate sensitivity for this crop, without providing evidence as statistically robust as that for corn.

For rice, sugarcane and cassava, the estimated coefficients do not indicate an average effect that is statistically different from zero. However, this lack of statistical significance should not be interpreted as evidence that these crops are insensitive to salinity. It may reflect limited exposure in the areas where they are grown, adaptations in agricultural practices, significant heterogeneity among cropping systems, or measurement error in salinity.

As explained above, an additional challenge comes from the design of the primary metric. Excess salinity is zero as long as electrical conductivity remains below the crop-specific threshold. This variable therefore contains a high proportion of zeros, particularly for the most tolerant crops, and captures only a portion of the available variation. For this reason, Table 13 uses mean salinity as an alternative measure.

Estimates based on mean salinity confirm the decline in corn yields, with comparable sign, order of magnitude, and precision. This convergence between two distinct definitions of exposure is the main source of robustness in the results. It shows that the estimated effect does not depend solely on the choice of agronomic thresholds or on the concentration of the main variable at zero.

The use of mean salinity also reveals a negative effect on soybeans. This result should, however, be interpreted with greater caution, since it is not replicated when using the measure of excess above the threshold. One possible explanation is that the tolerance threshold used for soybeans (5 dS/m) is rarely exceeded and therefore leaves few variations in the main specification that provide information.

The economic impact varies by crop. The most precise estimate concerns corn, which supplies both commercial markets and livestock feed chains. A decline in yield can therefore affect grain producers, as well as industries that use corn as an input (Silva, 2021). For beans and cassava, the lack of precise coefficients does not mean that the stakes are low. These crops occupy less of the national land area, but they play a significant role in the food supply and in family farming (Instituto Brasileiro de Geografia e Estatística, 2017; Wander and Silva, 2025; Empresa Brasileira de Pesquisa Agropecuária, 2025c). Finally, rice and sugarcane involve very different production systems. An average municipal coefficient may therefore mask responses specific to irrigated systems, upland fields, or agro-industrial chains. The data do not allow for the separation of these mechanisms in the main regression.

Overall, the results thus highlight a particularly pronounced vulnerability of corn to salinization, as well as potential, but less pronounced, effects on beans and soybeans. The heterogeneity among crops implies that salinization is not only an agronomic shock that uniformly reduces yields. It also affects the yields of different crops and, consequently, farmers' economic incentives. In the medium term, these differences in profitability may encourage a shift toward more tolerant crops, changes in irrigation practices, drainage adjustments, a reduction in cultivated areas, or the abandonment of certain crops.

This section focuses on observations by municipality and crop that were already in production in $t-1$. The planted area is positive in t and production is positive in $t-1$. Production in t is not used as a selection criteria. The planted area measures a land-use decision made before harvest. The quantity produced is then a combination of the yield and the area actually harvested. The analysis focuses on the variation at the municipal level between t and $t+1$.

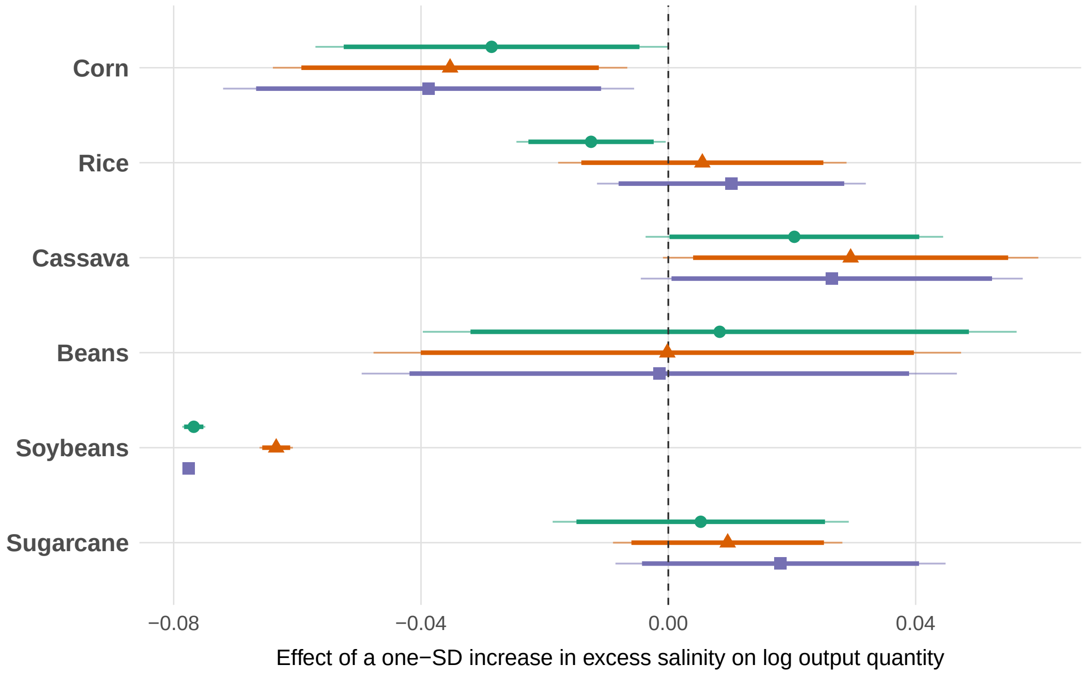
3.2 Soil Salinity and Crop Output

This subsection presents the contemporary SFD of equation (4), replacing the logarithm of the yield with the logarithm of the physical quantity produced. It therefore does not measure the variation between two growing seasons used for the planted area.

Figure 6 shows the coefficients on a common standardized scale. For corn, the preferred specification yields an effect of -0.035 log-points. A difference of one standard deviation in excess salinity, 0.164 dS/m in this sample, is thus associated with approximately 3.5% lower physical output. The coefficient is statistically significant. This result complements the decline in yield observed for corn. It indicates that the impact is reflected in the total quantity produced, without allowing for an exact distinction between yield and harvested area.

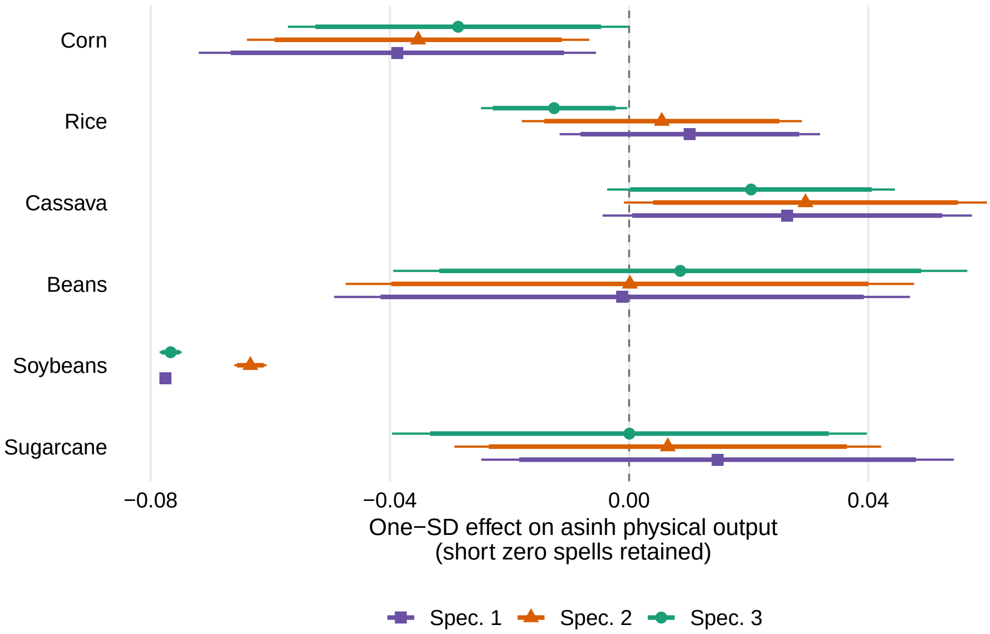
For cassava, the standardized effect is positive and close to 3.0%. It is significant at

Figure 6: Crop Output Responses to Excess Soil Salinity



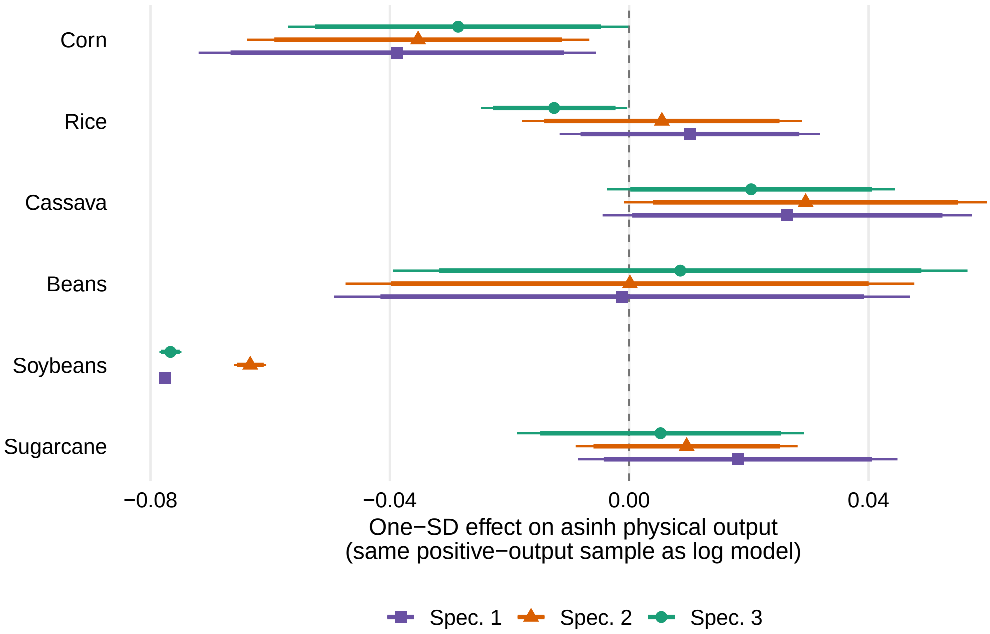
Notes: Points report crop-specific Spatial First-Difference (SFD) estimates at the municipality-pair-year level over 1985–2018. The dependent variable is the east-minus-west difference in log physical output quantity, conditional on strictly positive output in both municipalities. The treatment is the corresponding east-minus-west difference in crop-specific excess salinity above the FAO agronomic threshold, measured in dS/m. To facilitate comparisons across crops, coefficients and confidence intervals are multiplied by the crop-specific standard deviation of the treatment. Purple squares show Spec. 1 with year fixed effects only; orange triangles show preferred Spec. 2, which adds GDD, KDD and soil moisture; green circles show Spec. 3, which additionally controls for elevation, slope and clay content. Thick and thin lines report 90% and 95% confidence intervals based on Conley spatial standard errors with a 200 km cutoff. The three specifications use the same crop-specific estimation sample.

Figure 7: Crop Output Responses in Asinh with Short Zero Spells Retained



Notes: The dependent variable is the SFD in asinh physical output. Zero-output observations are retained when they belong to spells shorter than five consecutive years; longer zero spells are treated as structural missing values. Coefficients and confidence intervals are multiplied by the crop-specific treatment standard deviation. Purple squares show Spec. 1 with year fixed effects, orange triangles show preferred Spec. 2 with GDD, KDD and soil moisture, and green circles show Spec. 3 with additional elevation, slope and clay controls. Thick and thin lines report 90% and 95% confidence intervals based on Conley standard errors with a 200 km cutoff.

Figure 8: Crop Output Responses in Asinh on the Positive-Output Sample



Notes: The dependent variable is the SFD in asinh physical output, estimated on exactly the same crop-specific positive-output samples as the log model in Figure 6. The specifications, standardization and confidence intervals based on Conley spatial standard errors are otherwise identical to those in Figure 7. This comparison isolates the functional-form difference between log and asinh without changing the estimation sample.

the 10% level but not at the 5% level. This result does not indicate a reduction in yield and should not be presented as a positive response caused by salinity. The coefficients for rice, beans, and sugarcane are not statistically significant in the preferred specification.

The standardized coefficient for soybeans is negative and equals −0.063. However, it remains difficult to interpret. The standard deviation of the excess is only 0.015 dS/m, and the Conley covariance matrices require numerical correction in all three specifications. Standardization makes the magnitude comparable to that of other crops, but it does not introduce additional variation and does not resolve the inference problem. I therefore include the coefficient in the figure without treating it as a main result. The raw coefficients, treatment standard deviations, controls, and observations are presented in Table 22 in the appendix.

Figure 7 extends the analysis to episodes of zero output lasting fewer than five consecutive years. It therefore measures both variations in positive output and short-term transitions to or from zero. For corn, the estimated coefficient remains −0.035 and is significant at the 5% level. The results for rice, cassava, beans, and soybeans are also virtually unchanged. This similarity stems mainly from the fact that the expansion adds very few observations for these crops. It therefore does not allow us to conclude that transitions to zero are economically insignificant.

Sugarcane provides the most informative comparison. Its sample size increases from

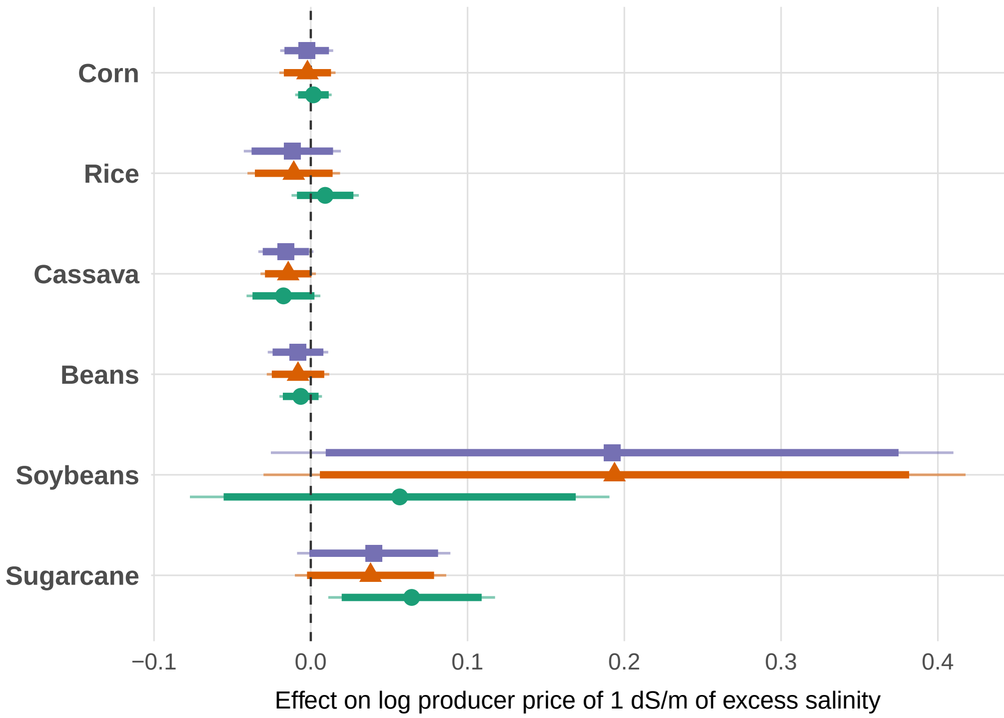
57,069 observations in the logarithmic model to 60,529 when short zero episodes are included. The standardized effect of the preferred specification decreases from 0.010 to 0.006 and becomes less precise. It remains insignificant. In this case, including zeros therefore weakens the estimated association among positive observations alone, rather than revealing an additional loss of production.

Figure 8 then applies asinh to exactly the same positive sample as the logarithm. The points and intervals almost coincide with those in Figure 6 across all three specifications. The difference between the logarithm and asinh is therefore negligible when the output is positive. The discrepancies observed in the figure with zeros stem from the change in sample and transitions involving zero, not from the transformation itself. When a transition involves zero, however, the coefficient in asinh should not be interpreted as an exact percentage. Table 23 summarizes the corresponding values in the appendix.

3.3 Crop Producer Prices

The decline in yield alone is not sufficient to understand the economic impact of salinity. An increase in the price received by the producer could offset part of the physical loss. The price structure and specifications are presented in Section 2.2.6. Figure 3 illustrates the national trend in real prices. The figure below addresses a different question: it tests whether local prices differ between two neighboring municipalities when their excess salinity levels differ.

Figure 9: Price Responses to Excess Soil Salinity



Notes: Points report SFD estimates of the effect of a 1 dS/m increase in crop-specific excess salinity on the log PAM average producer price. Green circles show Spec. 1 with year fixed effects and weather, soil and topographic controls. Orange triangles show Spec. 2 with pair and year fixed effects and weather controls. Purple squares show preferred Spec. 3 with pair and state-by-year fixed effects and weather controls. Thick and thin lines show 90% and 95% confidence intervals based on Conley spatial standard errors with a 200 km cutoff.

The estimates do not show robust compensation through producer prices. In the preferred specification, no coefficient is statistically significant at the 5% level. However, two associations emerge at the 10% significance level. Excess salinity is associated with a lower municipal price for cassava and a higher price for soybeans. The data therefore do not show that local prices consistently compensate for the physical losses associated with salinity.

This lack of a clear response does not mean that incomes remain unchanged. Gross income combines price, quantity, and area. It merely indicates that relative price variations between neighboring municipalities do not constitute a clearly identified compensation mechanism. This could be explained by the fact that there is still sufficient room for adaptation to cope with salinity.

The PAM does not allow us to distinguish between a price variation and a change in quality or market outlet. Price analysis therefore remains a complementary finding.

3.4 Historical Exposure to Soil Salinity and Yield Sensitivity

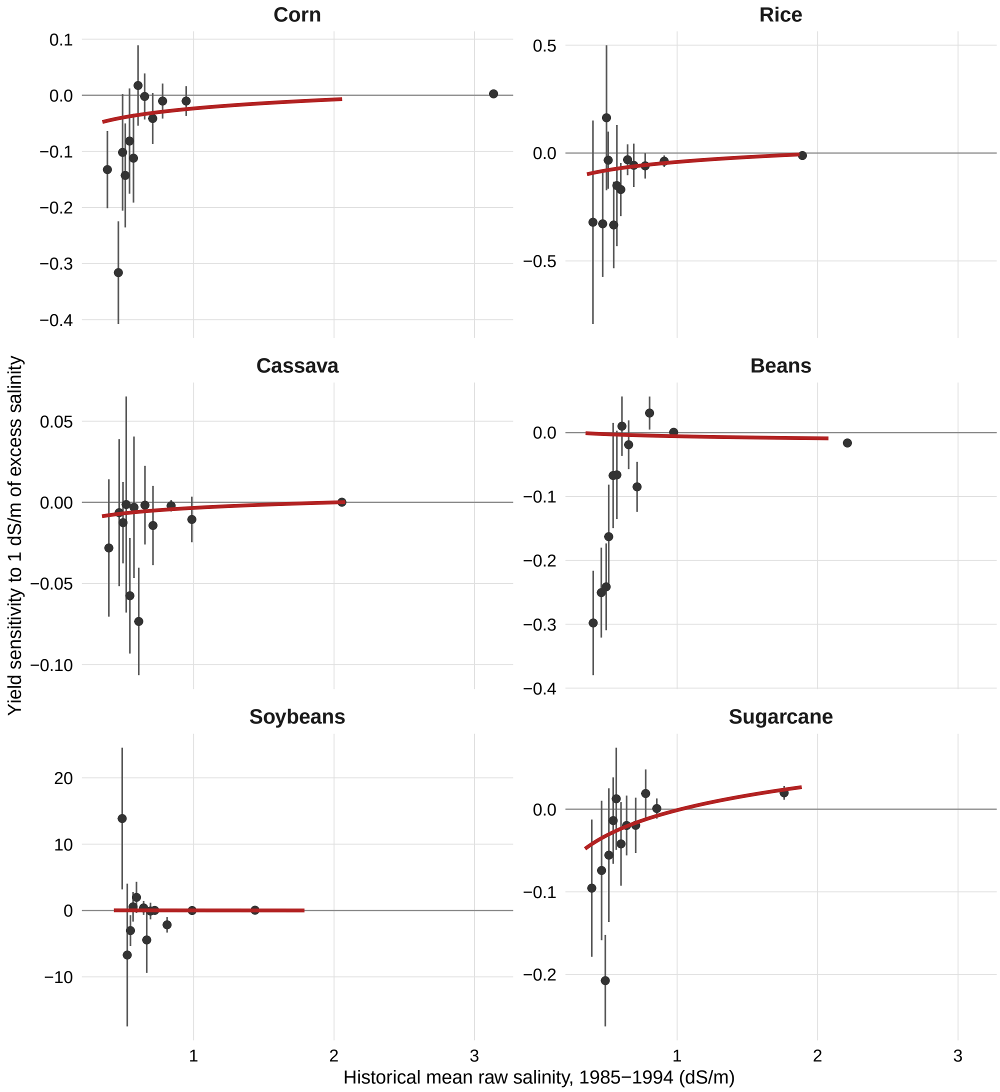
In the first step, I estimate how the yield of each crop responds to year-to-year deviations in excess salinity from the municipality's own average. This produces a municipality-specific measure of yield sensitivity to additional salinity stress. In the second step, following Butler and Huybers (2013) and Auffhammer (2022), I test whether this estimated sensitivity varies systematically with the municipality's historical salinity exposure. If municipalities that have historically faced higher salinity are less adversely affected by an additional salinity anomaly, this pattern is consistent with adaptation.

Figure 10 shows that the second-stage coefficients are positive and statistically significant for corn, rice, cassava, and sugarcane. They are 0.023, 0.055, 0.005, and 0.044, respectively. These signs indicate that the estimated sensitivity to excess salinity is less adverse in municipalities with historically higher salinity levels. The result is therefore consistent with gradual adaptation for the observed crops.

Beans do not show this adaptation pattern. The soybean coefficient is also non-significant, but its statistical power is very low: only 322 municipal sensitivities are estimated, and the median standard deviation of excess salinity within a municipality is only 0.0009 dS/m. The lack of results therefore does not allow us to conclude that there is no adaptation for this crop.

These results provide evidence consistent with adaptation, but not a causal measure of its mechanism. The data do not allow us to determine whether the attenuation stems from varieties, irrigation, drainage, other inputs, or the selection of land still under cultivation. Bean cultivation is primarily for domestic use and is carried out by smaller, family-run farms, but the data do not allow us to test whether this explains the absence of an attenuation estimate. Table 29 presents the detailed results.

Figure 10: Historical Salinity and Municipal Yield Sensitivity

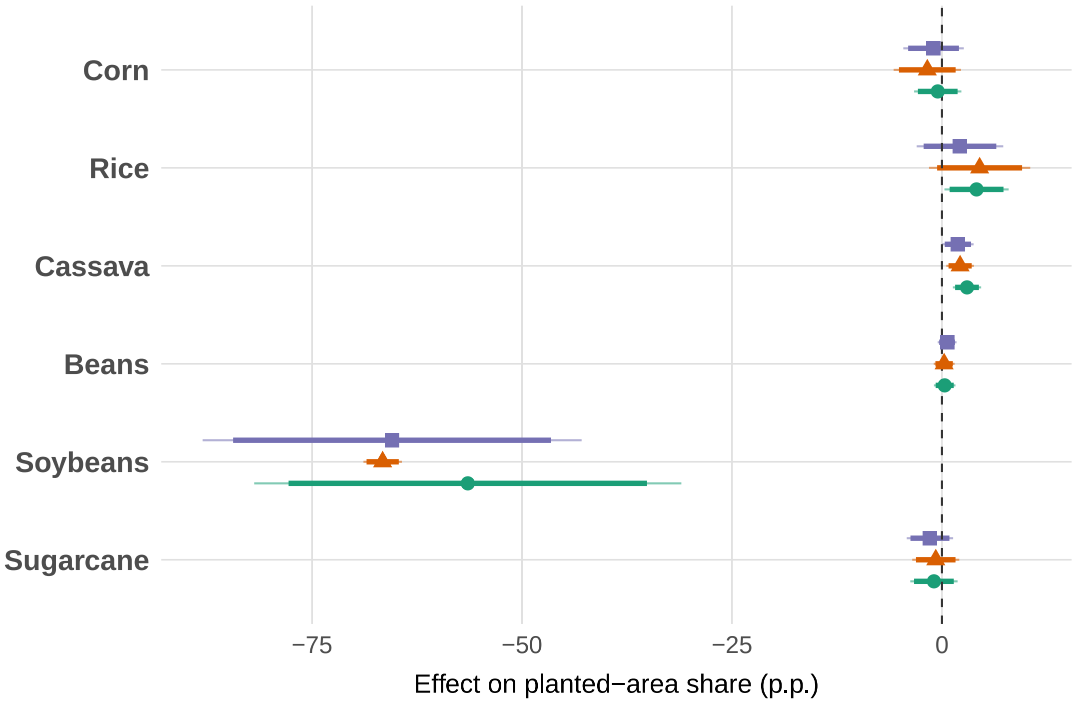


Notes: The figure follows the two-stage approach of Butler and Huybers (2013). Stage 1 estimates a separate 1985-2018 municipal sensitivity of log crop yield to annual crop-specific excess-salinity anomalies, controlling for a municipality-specific linear trend and GDD, KDD and soil-moisture anomalies. Each point is the inverse-variance-weighted mean sensitivity within one of twelve equally populated historical-salinity bins; bars show 95% intervals based on the first-stage bootstrap variances. Red lines are the logarithmic second-stage functions estimated on all individual municipal sensitivities. Historical salinity is the 1985-1994 municipal mean. Stage-1 uncertainty uses 199 circular moving-block residual bootstrap replications with three-year blocks.

3.5 Soil Salinity and Crop-Specific Planted Area

To complete the analysis of crop-specific responses, I first examine whether the salinity observed prior to the growing season is associated with a change in the composition of the planted areas within the municipality.

Figure 11: Municipal Crop Mix Responses to Lagged Excess Soil Salinity.



Notes: Points report Spatial First-Difference estimates of the effect of a 1 dS/m increase in lagged excess salinity on each crop's share of municipal planted area, in percentage points. Zeros in planted area are dropped only when they belong to a run of at least five consecutive zero years for the same municipality and crop. Green circles show Spec. 1 with year fixed effects only; orange triangles show Spec. 2, which adds lagged GDD, lagged KDD and lagged soil moisture; purple squares show Spec. 3, which additionally adds elevation, slope and clay. Thick lines show 90% confidence intervals and thin lines show 95% confidence intervals based on Conley standard errors with a 200 km cutoff.

The planted-area-share results in Figure 11 show that soybeans respond strongly to salinity exposure. Lagged excess salinity is associated with a negative and statistically significant change in the soybean share of municipal planted area in all three specifications. The coefficient is statistically significant at the 1 percent level in each specification.

The cassava share increases and remains statistically significant across the three specifications. For rice, the positive coefficient is significant only in the first specification, while the estimates for corn, beans, and sugarcane are statistically indistinguishable from zero. The joint decline in the soybean share and increase in the cassava share is consistent with a change in crop composition. However, the separate crop regressions do not establish that land is transferred directly from soybeans to cassava.

It might be surprising that the crops with the lowest thresholds are not the ones that appear to be most affected by a decline in planted area. However, the FAO thresholds reflect physiological tolerance, whereas this result reflects responses in terms of crop allocation. Consequently, the order of magnitude of responses in terms of planted area does not necessarily correspond to that of agronomic tolerance. This comparison proves particularly difficult for soybeans, as the 5 dS/m threshold is rarely exceeded in the estimation sample. The coefficient corresponding to each additional dS/m of excess salinity is therefore estimated based on a very limited number of treatments. Prices should not account for the results. In fact, even if I do not include prices in the specification, the SFD eliminates most of the price effect, and the annual crop-specific fixed effects absorb national crop price trends. What remains a potential source of confusion are differential local price shocks between neighboring municipalities.

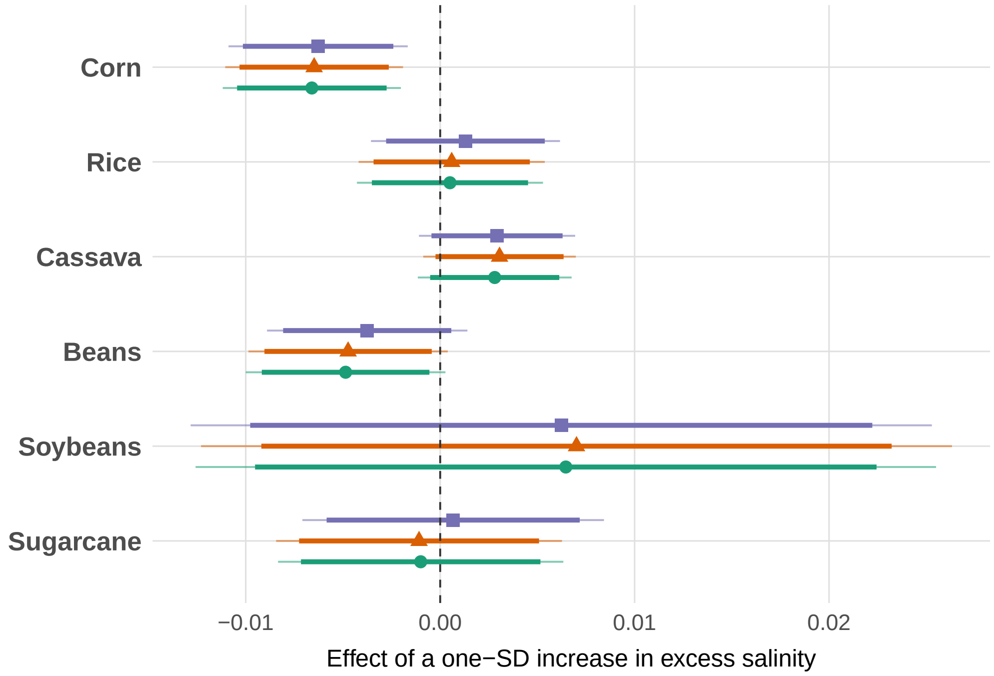
Since planted-area shares are relative outcomes, the negative soybean coefficient does not by itself reveal whether soybean acreage contracts in absolute terms. It may reflect a reduction in land planted with soybeans, an expansion of land allocated to other crops, or a combination of both. Moreover, municipality-level data do not allow us to observe whether individual farmers actively switch crops. Nevertheless, the results suggest that salinization affects the composition of local agricultural activity through the land-allocation margin.

The estimation results shown in Figure 12 address how soil salinity affects the area allocated to each crop in municipalities where that crop is already present. This is a response of the area used as a crop-specific input. It may reflect a reduction, compensation, or reallocation to other crops. The outcome under study is therefore the change in the municipal area devoted to a particular crop between t and $t+1$.

Figure 12 shows the effects of a one-standard-deviation increase in the excess salinity specific to each crop. This standardization makes it possible to compare crops on the same scale. For corn, the standard deviation of the treatment is 0.166 dS/m. When planted areas are positive and sufficiently large in both periods, this corresponds approximately to a 0.65% lower relative change in municipal corn planted area. The raw coefficient reported in Appendix Table 30 is -0.0391. Thus, a 1 dS/m increase in excess salinity is associated with a relative change in municipal corn planted area that is approximately 3.9% lower between t and $t+1$.

The results for other crops are different. In the main regression model, the coefficients for cassava and beans are not statistically significant at the 5% level. For beans, the coefficient is negative in the first two specifications, but becomes smaller after adding terrain controls. For rice, soybeans, and sugarcane, I also find no statistically significant effect on the variation in planted area. Corn is thus the only crop for which the result is clear in this section: the coefficient remains negative, similar, and statistically significant across all three specifications. Regressions are estimated separately for each crop. A decrease in the area planted with corn may therefore correspond to a switch to another

Figure 12: Excess Salinity and Change in Planted Area



Notes: An observation is one spatial municipality pair, crop and baseline campaign t. Baseline campaigns cover 1988-2017 and outcomes cover 1989-2018. In both municipalities, planted area is positive at t and production quantity is positive at $t-1$. The dependent variable is the east-minus-west difference in $\text{asinh}(A_{t+1}) - \text{asinh}(A_t)$. Planted areas equal to zero at $t+1$ are retained. For such a transition, the within-municipality change equals $-\text{asinh}(A_t)$ and does not have a percentage interpretation. The treatment is crop-specific excess salinity at t, measured in dS/m. To display all crops on a common scale, each coefficient and confidence interval is multiplied by the crop-specific sample standard deviation of the treatment. Green circles show Spec. 1 with year fixed effects. Orange triangles show preferred Spec. 2, which adds GDD, KDD and soil moisture at t. Purple squares show Spec. 3, which additionally adds elevation, slope and clay. Thick lines show 90% confidence intervals and thin lines show 95% confidence intervals based on standard errors clustered by spatial pair. The three specifications use the same crop-specific sample and contain no pair or state fixed effects. Raw coefficients per 1 dS/m and treatment standard deviations are reported in the appendix.

crop, to land set aside, or to land taken out of agricultural production. Zero areas in $t+1$ are retained, so the coefficient combines reductions in areas that remain positive with transitions to a zero municipal area.

The economic interpretation is consistent with the result in Section 3.1. An expected decline in corn yields reduces expected income per hectare. A higher risk of loss may also lead to planting less corn in the following growing season. Both explanations are possible. The result shows an adjustment in the municipal corn planted area.It does not show either a general shift in crops or a decline in total agricultural land.

3.6 Soil Salinity and Agricultural Land Use

This analysis complements Section 3.2.1, which focuses on the area of an existing crop. A decline in corn production can be compensated by an increase in soybean production without reducing the total cultivated area. Here, the outcome includes all crops. It measures a net change in the total cultivated area. The method is presented in Section 2.2.4.

3.6.1 Net Cropland Contraction

Table 4 shows a negative coefficient in all three specifications. In the preferred specification, when one municipality has average salinity 1 dS/m higher than its paired neighbor, its annual change in cultivated share is 0.065 percentage points lower. This magnitude represents approximately 0.47% of the average cultivated share in the sample. The coefficient remains statistically significant when the cutoff used to compute the Conley spatial standard errors varies from 50 to 300 km.

The result indicates a net relative contraction. New land classified as cropland automatically offsets land leaving this class. It therefore does not identify entries and exits separately. It also does not imply that the cultivated area declines in every municipality: the municipality with higher average salinity may lose more land or gain less land.

3.6.2 Timing and Economic Interpretation of Cropland Change

This subsection addresses a simple question: Does a municipality with higher salinity experience a less favorable trend in its total cultivated area starting the following year, or is this difference related to salinity that has been present for a longer period? I define t as the starting year. The result is the net change in the proportion of cropland between t and $t+1$. Salinity in t, $t-1$, and back to $t-5$ are simultaneously included in equation (11).

Panel A in Figure 13 breaks down the history of salinity. Each data point shows the association specific to a given year, with the other five years held constant. It is therefore used to identify the dates for which the association is most pronounced. Panel B answers

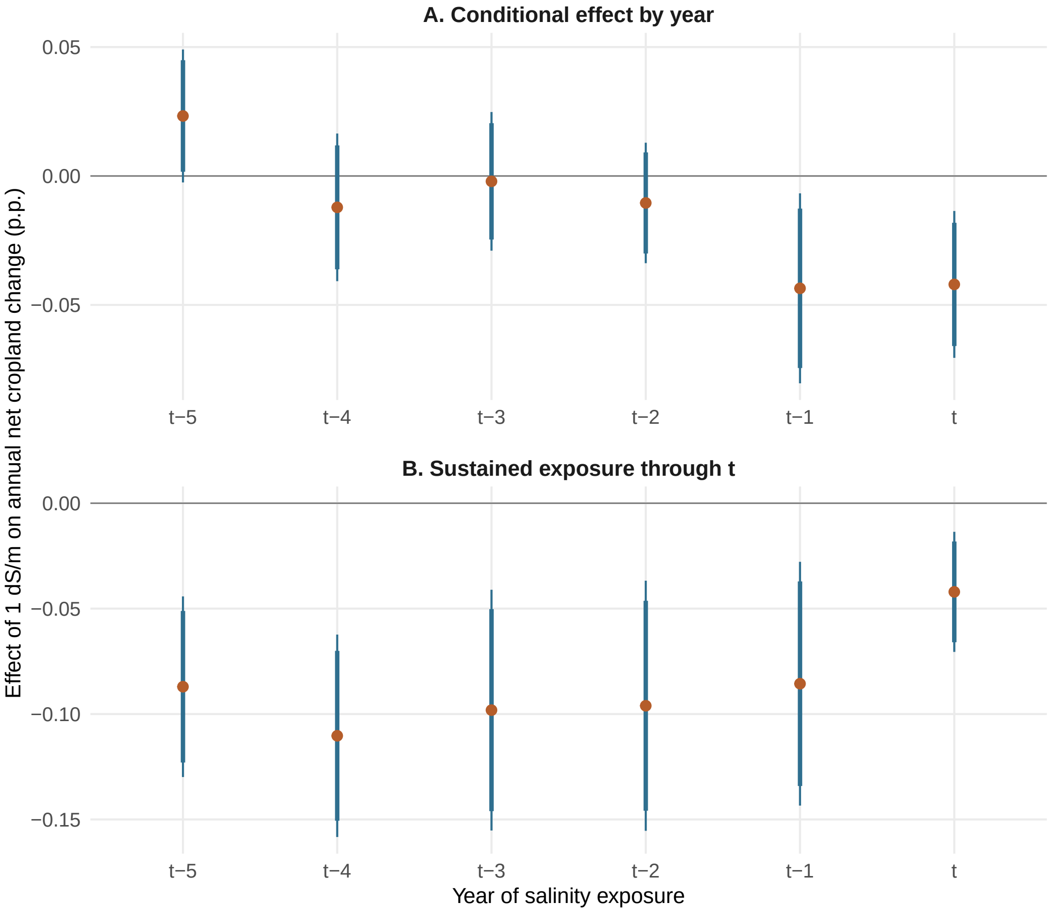
Table 4: Mean Salinity and the Annual Change in Total Physical Cropland

Specification	Year FE	Weather	Weather + soil/topography
Model:	(1)	(2)	(3)
Variables			
Δ_s Mean salinity at baseline (dS/m)	-0.0662***	-0.0649***	-0.0632***
	(0.0171)	(0.0161)	(0.0140)
Δ_s GDD at baseline		-3.34×10^{-5}*	-2×10^{-5}
		(2×10^{-5})	(2.03×10^{-5})
Δ_s KDD at baseline		-0.0005*	-0.0003
		(0.0003)	(0.0003)
Δ_s Soil moisture at baseline		0.1841	0.1809
		(0.2490)	(0.2566)
Δ_s Elevation			0.0002**
			(9.19×10^{-5})
Δ_s Slope			-0.0561***
			(0.0126)
Δ_s Clay			0.0042
			(0.0033)
Fixed-effects			
Year	Yes	Yes	Yes
Fit statistics			
R^2	0.00069	0.00137	0.00495
Observations	111,934	111,934	111,934

Conley (200km) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: An observation is a strictly contiguous east-west municipality pair and baseline year. The dependent variable is the east-minus-west difference in the annual percentage-point change in the share of municipal territory classified by MapBiomas as Agriculture (all crops combined, excluding pasture). Baseline years run from 1985 to 2017 and changes are measured from t to t+1. Zero cropland areas are retained. Mean salinity uses the same static historically cultivated municipal pixels as the other common-salinity specifications. Spec. 1 includes year fixed effects; Spec. 2 adds equally weighted crop-calendar GDD, KDD and soil moisture; Spec. 3 adds elevation, slope and clay. Spec. 2 is preferred because the SFD already absorbs much of the slowly varying soil and topographic heterogeneity shared by contiguous municipalities. All columns use the same complete-data sample. Standard errors are Conley spatial with a 200 km cutoff.

Figure 13: Timing of Relative Net Cropland Change in a Joint Distributed-Lag Model



Notes: Both panels are derived from the same preferred Spatial First-Difference regression; Panel B is not a second regression. The dependent variable is the east-minus-west difference in the annual percentage-point change in total physical cropland from t to $t+1$. A negative coefficient means that the municipality with higher average salinity has a lower annual net change in cropland share than its paired neighbor. Mean salinity at $t-5, t-4, \ldots, t$, measured over the common static set of historically cultivated municipal pixels, enters jointly. Panel A reports the coefficient for each salinity year conditional on the other five years. Panel B sums the coefficients from the date shown through t. For example, its point at $t-2$ sums the coefficients at $t-2$, $t-1$, and t, and therefore represents a 1 dS/m salinity difference sustained over these three dates. Its standard error is obtained by the delta method using the full Conley covariance matrix of the summed coefficients. These sums describe one annual cropland change, not accumulated land losses. The common sample contains 90,506 pair-year observations over 1990-2017. The model includes year fixed effects and GDD, KDD and soil moisture in t, but no pair fixed effects. Thick and thin lines show 90% and 95% confidence intervals based on Conley spatial standard errors with a 200 km cutoff.

another question. It sums the coefficients from the indicated date up to t. Its data point at $t-2$, for example, measures the association with a salinity difference of 1 dS/m held constant at $t-2$, $t-1$, and t. It is better suited to a persistent constraint than the isolated interpretation of a single year. Both panels use exactly the same coefficients estimated in the same regression.

In Panel A, the coefficients at t and $t-1$ are negative and statistically significant. They equal -0.042 and -0.044 percentage points for a 1 dS/m difference. The coefficients from $t-2$ to $t-4$ are not statistically significant. The coefficient at $t-5$ is positive and significant only at the 10% level. I therefore do not interpret this isolated coefficient as evidence of a delayed beneficial response. Salinity changes slowly and the six values are correlated, so each isolated coefficient must be read conditionally on the other five.

Panel B is better suited to a persistent difference in salinity. A difference of 1 dS/m maintained at $t-2$, $t-1$, and t is associated with an annual variation in the cultivated share that is 0.096 percentage points lower in the municipality with higher average salinity than in its paired neighbor. This value represents approximately 0.70% of the average cultivated share, which is 13.68%. For a municipality covering 100 000 hectares, 0.096 percentage points corresponds to 96 hectares. This is only an illustration because municipal areas differ.

The sum of all six coefficients is -0.087 percentage points and remains statistically significant. It measures one annual change associated with a salinity difference maintained over the six dates.

The result applies to all crops combined. A simple shift from corn to soybeans, or from another crop to corn, does not reduce this measure if the same land remains under cultivation. The negative coefficient indicates a significantly smaller variation in the municipalities most affected by soil salinity. It does not, on its own, show that these municipalities are losing land in absolute terms. The discrepancy may stem from a greater loss of land from the cultivated land category, a smaller increase in land entering agricultural use, or both. It may also reflect land being set aside, conversion to pasture or another use, or a slower rate of opening up new cultivated land. The data do not allow us to distinguish between these mechanisms.

This analysis complements that in Section 3.2.1. The latter measures the reduction in the area devoted to a crop that was already present. This reduction may be offset by the expansion of another crop and therefore does not indicate an overall decline in agricultural production. Here, the comparison pertains to the variation in total physical area under cultivation. The result thus suggests that part of the adjustment associated with salinity goes beyond the reallocation between crops. It corresponds to less land being used for crops in areas with higher salinity.

Another issue may be the conversion of these lands to activities that are more compatible with high salinity. Appendix Table 34 examines municipal aquaculture areas. However, the aquaculture sector, although growing in Brazil, is not yet a major sector.

3.7 Soil salinity and migration

In this section, I present results on the extensive margin of adaptation with respect to migration. Table 5 reports the PPML gravity specification defined in equation (14). The dependent variable is a bilateral migrant flow, while the origin-population offset gives its conditional mean a rate interpretation.

Table 5: Soil Salinity and Municipal Migration from Rural Origins

Dependent Variable:	Migrant flow (origin-population offset)	
Model:	(1)	(2)
Variables		
Origin mean salinity (std.)	-0.0141	-0.0124
	(0.0152)	(0.0144)
Destination mean salinity (std.)	0.0061	0.0030
	(0.0227)	(0.0214)
Origin GDD (std.)		0.0154
		(0.0297)
Origin KDD (std.)		0.0876*
		(0.0521)
Origin soil moisture (std.)		0.1184***
		(0.0407)
Destination GDD (std.)		0.0072
		(0.0259)
Destination KDD (std.)		0.1075**
		(0.0429)
Destination soil moisture (std.)		0.0687
		(0.0435)
Fixed-effects		
Dyad fixed effects	Yes	Yes
Origin-state-by-window fixed effects	Yes	Yes
Destination-state-by-window fixed effects	Yes	Yes
Fit statistics		
Observations	111,310	111,310
Squared Correlation	0.88048	0.88199
Pseudo R^2	0.79763	0.79810

Clustered (orig & dest) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: PPML estimates of weighted bilateral migration flows for people aged 15 or older. The flow records the last intermunicipal move and is assigned to a five-year window from the reported duration in the current municipality. Origins are rural when at least 50 percent of their weighted population lived in rural areas in the baseline census; destinations are unrestricted. The estimated windows are 1986–1990, 2001–2005 and 2006–2010. Salinity, GDD, KDD and soil moisture are measured in 1985, 2000 and 2005, respectively, before the corresponding migration window, and standardized within window. Both columns include dyad, origin-state-by-window and destination-state-by-window fixed effects. Column 2 is preferred and adds GDD, KDD and soil moisture at origin and destination; precipitation and income are excluded. The offset is the weighted population aged 15 or older in the origin baseline census. All columns use the same final sample; standard errors are two-way clustered by origin and destination.

The table uses the last-move construction described in Section 2.1.4. Four windows are reconstructed, but only three enter the regression with salinity measured before migration. Both columns use the same 111,310 observations, including 55,998 zero-flow cells, 37,141 retained dyads, 441 rural origins, and 1,802 destinations. Each retained dyad records at least one positive flow during an estimated window; dyads that remain zero in every window are outside the estimation panel.

In the first column, the coefficient on salinity at origin is -0.0141. It remains close to zero after climate controls are added. In the preferred specification, it is -0.0124, with a standard error of 0.0144. For a retained dyad, a one-standard-deviation increase in origin salinity is associated with an expected bilateral out-migration rate about 1.2% lower. The coefficient is not statistically significant. Salinity at destination is not statistically significant either.

The negative sign is therefore not sufficient to conclude that salinity reduces departures. The confidence interval remains consistent with either a modest decrease or a modest increase in the expected bilateral rate. The result supports neither salinity-induced out-migration nor constrained immobility. It concerns all individuals observed from rural origins and does not identify farmers at departure.

Dyad fixed effects absorb distance, persistent migration networks, and other constant bilateral characteristics. The stability of the coefficient after adding climate controls only indicates that these controls do not explain its sign. The estimate remains a conditional association. This caution is consistent with the literature, which shows that migration may respond differently to rapid shocks and to slow environmental degradation (Feng et al., 2010; Jessoe et al., 2018; Branco and Feres, 2021).

Heterogeneity by income quartile. I expand on this analysis by examining the heterogeneity of migration by income quartile. Table 6 divides origins into income quartiles within each migration time window and re-estimates the preferred PPML specification separately in each quartile. This avoids comparing nominal income levels across censuses and currencies. The association with salinity at origin is positive in the poorest quartile and negative in the richest quartile. The corresponding changes in the expected bilateral migration flows are approximately +4.0% in first-quartile and -3.7% in fourth-quartile. Both coefficients are statistically significant only at the 10% level. The second-quartile and third-quartile coefficients are not statistically significant, and the pattern is not monotone across income groups. These results therefore do not support a simple poverty-trap mechanism in which lower-income rural residents are systematically less able to leave more saline origins. The positive association for the poorest origins is nevertheless consistent with the possibility that limited access to in-situ adaptation options makes migration a relatively more important response to environmental degradation. Conversely, the negative coefficient among the richest origins could be consistent with greater capacity to adapt locally. However, the data do not directly measure access to adaptation technologies or household resources at the destination and at the origin location, and differences across income quartiles are not formal tests of coefficient heterogeneity. These mechanisms therefore remain suggestive rather than identified.

Table 6: Migration from Rural Origins: Separate Regressions by Baseline-Income Quartile

Variable	Q1 (poorest)	Q2	Q3	Q4 (richest)
Origin mean salinity (std.)	0.0397*	0.0040	-0.0784	-0.0378*
	(0.0213)	(0.0193)	(0.0595)	(0.0202)
Destination mean salinity (std.)	-0.0256	-0.0222	0.0325	-0.0005
	(0.0305)	(0.0496)	(0.0315)	(0.0306)
Origin GDD (std.)	0.0294	0.0877*	-0.0031	0.1654
	(0.0575)	(0.0529)	(0.0483)	(0.1155)
Origin KDD (std.)	0.0026	0.1144	0.0674	0.2377
	(0.0460)	(0.0741)	(0.0601)	(0.1577)
Origin soil moisture (std.)	0.0702	0.0375	0.1084	0.1021
	(0.0567)	(0.0586)	(0.0795)	(0.0915)
Destination GDD (std.)	-0.0164	-0.0165	-0.0208	0.1033**
	(0.0593)	(0.0517)	(0.0366)	(0.0493)
Destination KDD (std.)	0.0193	0.1080*	0.0777	0.1018
	(0.0606)	(0.0617)	(0.0562)	(0.0720)
Destination soil moisture (std.)	0.0780	0.1017**	0.0101	0.0627
	(0.0624)	(0.0448)	(0.0800)	(0.0720)
Dyad fixed effects	Yes	Yes	Yes	Yes
Origin-state-by-window fixed effects	Yes	Yes	Yes	Yes
Destination-state-by-window fixed effects	Yes	Yes	Yes	Yes
Observations	19,041	18,459	23,008	31,079
Origins	110	110	110	111
Dyads	7,410	7,638	8,739	10,813
Origin-windows	277	262	282	315
Mean origin salinity (std.)	-0.331	-0.382	-0.158	0.520
Squared correlation	0.921	0.899	0.903	0.894
Pseudo R2	0.810	0.803	0.798	0.828

Notes: Each column is a separate PPML regression estimated only on origin–destination–window observations in the indicated baseline-income quartile. Q1 is the poorest quartile and Q4 the richest; quartiles are formed within each migration window from mean origin income measured in its baseline census, avoiding comparisons of nominal income levels across currencies. The outcome, offset and specification are those of the preferred migration model. Every regression includes destination salinity, GDD, KDD and soil moisture at origin and destination, dyad fixed effects, origin-state-by-window fixed effects and destination-state-by-window fixed effects. Precipitation and income are excluded. Standard errors in parentheses are two-way clustered by origin and destination. PPML may remove fixed-effect groups with only zero flows; the observation counts therefore refer to the final estimation sample of each separate regression. Because the four models are estimated separately, differences between columns are descriptive and are not formal tests of coefficient equality.

4 Conclusion

This thesis examines the relationship between soil salinization and Brazilian agriculture from 1985 to 2018. The spatial first-differences strategy compares neighboring municipalities and accounts for meteorological conditions, soil characteristics, and topography. The main finding concerns corn. In the preferred specification, a 1 dS/m increase in excess salinity is associated with a decrease of approximately 7.8% in yield. This result remains valid when using average salinity. Corn production is also lower as excess salinity increases. For other crops, the coefficients are less precise or depend more heavily on the measure of exposure. Soybeans illustrate this limitation: their agronomic threshold is rarely exceeded, and the large estimated values lie far from the range actually observed. The comparison inspired by Butler and Huybers (2013) and Auffhammer (2022) provides a complementary result. Sensitivity to excess salinity becomes less adverse as historical exposure increases for corn, rice, cassava, and sugarcane. These results are consistent with gradual adaptation, though they do not allow us to identify the specific agricultural practice responsible for it. The results also show that yield does not fully account for agricultural adjustment. Across all crops combined, when one municipality has average salinity 1 dS/m higher than its paired neighbor, its annual change in cultivated share is 0.065 percentage points lower. Farm-gate prices show no robust response.

The main model reconstructs the most recent intermunicipal migration in four approximately five-year windows and uses the three windows for which salinity can be measured before migration. In the preferred specification, the coefficient on origin salinity is small, negative, and not statistically significant. It therefore provides no evidence that salinity changes the expected bilateral out-migration flows within the retained dyads or that it constrains mobility. The migration results require caution.

My findings have nevertheless several implications for public policy in Brazil. Salinity monitoring should not be separated from agricultural policies. The losses observed in corn production warrant closer monitoring in municipalities where the agronomic threshold has been exceeded. They also highlight the importance of technical assistance regarding water quality, drainage, and soil management practices. However, the variation across crops does not justify a one-size-fits-all recommendation to switch crops. Policies must take local production systems into account and verify that the proposed solutions actually reduce exposure. Finally, the lack of specific data on migration makes it impossible to view the departure as a spontaneous adaptation to soil salinization. Data on households, their occupations before and after migration, and their resources prior to displacement would be necessary to distinguish more precisely these mechanisms.

These limitations are important for interpretation. PAM data are aggregated at the municipal level and do not track individual plots or producers. They do not allow for distinguishing between the varieties used for a given crop. This limitation is particularly significant, as more tolerant varieties may respond differently to the same level of salinity.

The thesis also lacks a dataset of microdata on farms, their inputs, access to credit, irrigation technology, or the quality of the water used. Migration data do not allow for a precise assessment of a migrant's occupation and resources at their place of origin. Finally, the predicted maps and their municipal aggregation may smooth out very low and very high salinity levels. This smoothing reduces the contrasts measured between neighboring areas and may attenuate the coefficients. The estimated magnitudes may therefore be conservative, without constituting a formal lower bound.

Future research could link local electrical conductivity measurements to plot- and farm-level data. It could compare crop varieties, irrigation and drainage techniques, as well as access to credit and technical assistance. An individual-level panel would also make it possible to track changes in activity, income, and migration among rural households. These data would help distinguish between successful technical adaptation and a forced reduction in production or cultivated land. Finally, they would allow for a direct evaluation of salinity management policies and the adoption of salt-tolerant varieties.

A Data Construction and Sample Diagnostics

A.1 Spatial-pair restrictions

Table 7 reports the number of municipalities retained or excluded when constructing strictly contiguous pairs within the same state, hydrological basin, and latitude channel.

Table 7: Spatial-Pair Restriction Diagnostics

Metric	Value
Municipalities in PAM zero panel	5,425
West-east pairs retained	4,306
Municipalities with admissible western neighbor	4,306
Municipalities appearing on either side of a pair	4,676
Municipalities never appearing in a final pair	749
Municipalities without admissible western-neighbor role	1,119

Notes: Final pairs compare each municipality with its strictly contiguous western neighbor within the same state, main hydrological basin, and latitude channel. The number of excluded municipalities depends on the definition: some municipalities have no admissible western neighbor but appear as the western neighbor of another municipality.

Table 8: Within-Pair Variation in Salinity and Observed Covariates

Variable	Unit	Mean SFD	SD of SFD	Relative SD (%)
Raw mean ECe	dS/m	0.003	0.242	46.1
Crop-specific excess ECe	dS/m	0.003	0.167	48.1
GDD	degree-days	-4.365	250.300	16.9
KDD	degree-days	-0.719	21.716	20.1
Soil moisture	m3/m3	-0.000	0.025	24.7
Elevation	m	-2.730	109.002	28.0
Slope	%	0.015	0.360	24.3
Clay content	%	0.091	2.650	21.0

Notes: Statistics use the common 1985–2018 sample of 467,401 east-minus-west municipality-pair–crop–year observations entering the main log-yield SFD regressions. Relative SD is the SFD standard deviation divided by sqrt(2) times the standard deviation of the corresponding municipal levels; 100 percent is the benchmark for two independent draws from the same sample, so lower values indicate greater local similarity. The pairs are in the same state and main hydrological basin by construction. Municipal producer-price SFDs are reported in a separate crop-specific diagnostic table; a consistent municipal measure of market access is not available and is not claimed to be directly balanced.

A.2 Treatment of zero observations in PAM

Table 9 documents the balanced municipality–crop–year support obtained after completing the PAM panel with zeros. Table 10 then distinguishes short zero spells from zeros belonging to runs of at least five consecutive years. The latter are treated as structural absences in the zero-inclusive analyses, whereas shorter spells are retained through the inverse hyperbolic sine transformation.

Table 9: PAM Zero-Completed Panel Balance

Crop	Observations	Municipalities	Years	Missing yld.	Zero qty.	Zero planted
beans	184,450	5,425	34	13,988	20,594	19,011
cassava	184,450	5,425	34	13,988	24,175	22,446
corn	184,450	5,425	34	13,988	7,978	5,754
rice	184,450	5,425	34	13,988	59,536	56,026
soy	184,450	5,425	34	13,988	115,681	106,933
sugarcane	184,450	5,425	34	13,988	59,814	55,802

Notes: The zero-completed PAM support is balanced by construction: 5,425 municipalities, six crops and 34 years produce 1,106,700 municipality–crop–year cells. The log regressions are not balanced because yield and log production are undefined when a crop is not effectively produced.

Table 10: Five-Year Persistent-Zero Diagnostic

Crop	Cells	Zero qty.	Qty. zero 5yr	Zero planted	Planted zero 5yr
beans	184,450	20,594	14,108	19,011	13,761
cassava	184,450	24,175	20,212	22,446	18,773
corn	184,450	7,978	3,614	5,754	3,332
rice	184,450	59,536	53,296	56,026	50,730
soy	184,450	115,681	112,137	106,933	103,810
sugarcane	184,450	59,814	55,529	55,802	51,779

Notes: A zero is treated as structural only when it belongs to a run of at least five consecutive zero years for the same municipality and crop. Short zero spells are retained in asinh specifications; they cannot directly enter log specifications.

B Crop-Yield Results and Robustness

B.1 Nonlinear response to raw mean salinity

Table 11 reports the estimates and support counts underlying Figure 4. It makes explicit the limited identifying variation in the highest salinity bins.

Table 11: Detailed Nonlinear Yield Responses Across Raw Mean Salinity Bins

Panel A: Corn, rice, and cassava

Salinity bin (dS/m)	Corn Estimate (SE)	Corn Support	Rice Estimate (SE)	Rice Support	Cassava Estimate (SE)	Cassava Support
< 0.5	Reference	63,908	Reference	33,185	Reference	53,520
0.5-1.0	-0.0029 (0.0057)	23,788	-0.0015 (0.0056)	14,111	0.0020 (0.0028)	20,812
1.0-1.5	-0.0147 (0.0094)	7,039	0.0203 (0.0226)	4,388	0.0076 (0.0117)	6,260
1.5-2.0	-0.0302^{**} (0.0134)	2,639	0.0671^{*} (0.0378)	1,759	0.0110 (0.0135)	2,553
2.0-2.5	-0.0352^{**} (0.0154)	1,158	0.0501 (0.0533)	807	0.0241 (0.0149)	1,117
2.5-3.0	-0.1222^{***} (0.0376)	657	-0.0145 (0.0576)	467	-0.0162 (0.0216)	620
3.0-4.0	-0.1948^{***} (0.0538)	442	0.0440 (0.0465)	333	-0.1069 (0.0853)	421
4.0-5.0	Not estimated	104	Not estimated	71	Not estimated	105
≥ 5.0	Not estimated	35	Not estimated	26	Not estimated	36
Regression observations	114,025		66,072		94,296	

Panel B: Beans, soybeans, and sugarcane

Salinity bin (dS/m)	Beans Estimate (SE)	Beans Support	Soybeans Estimate (SE)	Soybeans Support	Sugarcane Estimate (SE)	Sugarcane Support
< 0.5	Reference	57,201	Reference	19,446	Reference	31,071
0.5-1.0	0.0070 (0.0050)	21,716	-0.0016 (0.0045)	8,035	0.0207^{***} (0.0077)	11,753
1.0-1.5	-0.0020 (0.0095)	6,732	-0.0061 (0.0137)	1,629	0.0258 (0.0188)	3,445
1.5-2.0	-0.0042 (0.0141)	2,539	-0.0168 (0.0149)	658	0.0366^{**} (0.0157)	1,266
2.0-2.5	-0.0130 (0.0299)	1,143	-0.0120 (0.0381)	261	0.0230 (0.0347)	612
2.5-3.0	-0.0584^{**} (0.0274)	621	Not estimated	120	0.0523 (0.0467)	321
3.0-4.0	-0.1026^{**} (0.0436)	433	Not estimated	92	Not estimated	196
4.0-5.0	Not estimated	102	Not estimated	4	Not estimated	49
≥ 5.0	Not estimated	35	Not estimated	0	Not estimated	10
Regression observations	102,376		33,563		57,962	

Notes: The table reports the coefficients underlying Figure 4. The dependent variable is the east-minus-west difference in log yield. Raw mean salinity below 0.5 dS/m is the omitted category. Models include year fixed effects and differences in GDD, KDD, soil moisture, slope, elevation, and clay content. Standard errors in parentheses use the Conley spatial estimator with a 200 km cutoff. Support reproduces the identifying count stored with each bin estimate; cells marked 'Not estimated' lack sufficient identifying variation in the exported model. *, **, and *** denote significance at the 10, 5, and 1 percent levels.

B.2 Detailed crop-yield regressions

The following tables report the complete estimates underlying the crop-yield results. They present the three specifications for crop-specific excess salinity and raw mean salinity, including all estimated controls and Conley standard errors with a 200 km cutoff.

Table 12: Crop Yields and Excess Salinity (Spatial Conley Correction, 200 km)

Panel A: Corn, rice, and cassava

Crop		Corn			Rice			Cassava	
Spec	Spec 1	Spec 2	Spec 3	Spec 1	Spec 2	Spec 3	Spec 1	Spec 2	Spec 3
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Δ_s Excess Salinity (Above FAO, dS/m)	-0.0848***	-0.0814***	-0.0732***	0.0423	0.0326	0.0119	-0.0232	-0.0227	-0.0227
	(0.0246)	(0.0233)	(0.0217)	(0.0619)	(0.0545)	(0.0439)	(0.0176)	(0.0173)	(0.0173)
Δ_s GDD (degree-days)		5.31×10^{-6}	6.19×10^{-5}*		0.0002***	1.21×10^{-5}		-4.35×10^{-6}	-3.95×10^{-6}
		(4.15×10^{-5})	(3.45×10^{-5})		(7.19×10^{-5})	(5.68×10^{-5})		(9.14×10^{-6})	(8.9×10^{-6})
Δ_s KDD (degree-days)		-0.0010***	-0.0007*		0.0086	0.0052		-2.93×10^{-5}	-1.66×10^{-5}
		(0.0004)	(0.0004)		(0.0063)	(0.0061)		(0.0001)	(0.0001)
Δ_s Soil Moisture (m^3/m^3)		0.3879*	0.3376*		-0.0084	-0.0296		0.1772*	0.1757*
		(0.2011)	(0.1902)		(0.4849)	(0.4188)		(0.0976)	(0.1061)
Δ_s Slope Gradient (%)			-0.0236***			-0.0232			-0.0050
			(0.0088)			(0.0186)			(0.0034)
Δ_s Elevation (m)			0.0002*			-0.0004**			1.45×10^{-5}
			(8.54×10^{-5})			(0.0002)			(4.96×10^{-5})
Δ_s Clay Content (%)			0.0076***			0.0079*			0.0007
			(0.0019)			(0.0045)			(0.0018)
R^2	0.00100	0.00361	0.00716	0.00026	0.00653	0.01189	0.00026	0.00050	0.00064
Observations	114,025	114,025	114,025	66,072	66,072	66,072	94,296	94,296	94,296
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Table 12 (continued)

Panel B: Beans, soybeans, and sugarcane

Crop	Beans			Soybeans			Sugarcane		
Spec	Spec 1	Spec 2	Spec 3	Spec 1	Spec 2	Spec 3	Spec 1	Spec 2	Spec 3
	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)
Δ_s Excess Salinity (Above FAO, dS/m)	-0.0300*	-0.0295**	-0.0245*	-0.1869	-0.1350	-0.1674	-0.0091	-0.0152	-0.0222
	(0.0154)	(0.0150)	(0.0140)	(0.2489)	(0.2243)	(0.2303)	(0.0122)	(0.0124)	(0.0148)
Δ_s GDD (degree-days)		-4.17×10^{-5}	1.85×10^{-5}		-0.0001*	6.12×10^{-5}		0.0002***	9.33×10^{-5}
		(4.87×10^{-5})	(3.73×10^{-5})		(6.36×10^{-5})	(7.68×10^{-5})		(7.28×10^{-5})	(6.63×10^{-5})
Δ_s KDD (degree-days)		-0.0003	-0.0002		-0.0001	-0.0004		-0.0006	-0.0011***
		(0.0004)	(0.0005)		(0.0002)	(0.0003)		(0.0004)	(0.0004)
Δ_s Soil Moisture (m^3/m^3)		0.1922	0.1704		0.0563	0.0819		-0.0035	-0.0110
		(0.1665)	(0.1649)		(0.1188)	(0.0834)		(0.2284)	(0.2403)
Δ_s Slope Gradient (%)			-0.0260***			-0.0524***			0.0046
			(0.0061)			(0.0140)			(0.0072)
Δ_s Elevation (m)			0.0002**			0.0002*			-0.0003***
			(7.04×10^{-5})			(9.75×10^{-5})			(6.68×10^{-5})
Δ_s Clay Content (%)			0.0036			0.0032**			0.0040*
			(0.0024)			(0.0013)			(0.0021)
R^2	0.00027	0.00093	0.00330	0.00061	0.00377	0.01913	0.00032	0.00337	0.00582
Observations	102,376	102,376	102,376	33,563	33,563	33,563	57,069	57,069	57,069
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The dependent variable is the spatial first difference of log crop yield. The treatment is the spatial difference in contemporaneous excess salinity above the crop-specific FAO threshold. All differences are east minus west. Pairs link west-neighbor municipalities within the same State, Hydrological Basin, and latitude channel. Spec. 1 includes year fixed effects; Spec. 2 adds GDD, KDD, and soil moisture; Spec. 3 additionally controls for slope, elevation, and clay content. Salinity and agro-climatic controls are measured over the contemporaneous growing season. A crop-specific common complete-case sample is used across specifications. Conley (1999) spatial HAC standard errors (200 km cutoff; spherical distance) are in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

Table 13: Crop Yields and Mean Soil Salinity (Spatial Conley Correction, 200 km)

Panel A: Corn, rice, and cassava

Crop		Corn			Rice			Cassava	
Spec	Spec 1	Spec 2	Spec 3	Spec 1	Spec 2	Spec 3	Spec 1	Spec 2	Spec 3
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Δ_s Mean Salinity (dS/m)	-0.0605***	-0.0604***	-0.0526***	0.0200	0.0164	0.0151	-0.0137	-0.0138	-0.0134
	(0.0151)	(0.0144)	(0.0129)	(0.0320)	(0.0300)	(0.0245)	(0.0133)	(0.0132)	(0.0129)
Δ_s GDD (degree-days)		5.6×10^{-6}	$6.28 \times 10^{-5*}$		0.0002***	1.31×10^{-5}		-4.32×10^{-6}	-3.9×10^{-6}
		(4.12×10^{-5})	(3.44×10^{-5})		(7.16×10^{-5})	(5.55×10^{-5})		(9.18×10^{-6})	(8.92×10^{-6})
Δ_s KDD (degree-days)		-0.0010***	-0.0007*		0.0086	0.0052		-3.27×10^{-5}	-1.69×10^{-5}
		(0.0004)	(0.0004)		(0.0064)	(0.0061)		(0.0001)	(0.0001)
Δ_s Soil Moisture (m^3/m^3)		0.3933**	0.3430*		-0.0134	-0.0339		0.1794*	0.1770*
		(0.2001)	(0.1895)		(0.4916)	(0.4234)		(0.0985)	(0.1069)
Δ_s Slope Gradient (%)			-0.0232***			-0.0232			-0.0048
			(0.0088)			(0.0187)			(0.0034)
Δ_s Elevation (m)			0.0002*			-0.0004**			1.58×10^{-5}
			(8.53×10^{-5})			(0.0002)			(4.94×10^{-5})
Δ_s Clay Content (%)			0.0075***			0.0080*			0.0007
			(0.0019)			(0.0044)			(0.0017)
R^2	0.00108	0.00376	0.00723	0.00028	0.00655	0.01195	0.00020	0.00045	0.00059
Observations	114,025	114,025	114,025	66,072	66,072	66,072	94,296	94,296	94,296
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Table 13 (continued)

Panel B: Beans, soybeans, and sugarcane

Crop	Beans			Soybeans			Sugarcane		
Spec	Spec 1	Spec 2	Spec 3	Spec 1	Spec 2	Spec 3	Spec 1	Spec 2	Spec 3
	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)
Δ_s Mean Salinity (dS/m)	-0.0239*	-0.0247*	-0.0200	-0.0697***	-0.0652***	-0.0422**	-0.0039	-0.0065	-0.0085
	(0.0138)	(0.0134)	(0.0123)	(0.0183)	(0.0165)	(0.0185)	(0.0098)	(0.0095)	(0.0098)
Δ_s GDD (degree-days)		-4.21×10^{-5}	1.87×10^{-5}		-9.94×10^{-5}	5.38×10^{-5}		0.0002***	9.45×10^{-5}
		(4.87×10^{-5})	(3.74×10^{-5})		(6.17×10^{-5})	(7.37×10^{-5})		(7.06×10^{-5})	(6.43×10^{-5})
Δ_s KDD (degree-days)		-0.0003	-0.0002		-9.29×10^{-5}	-0.0004		-0.0006	-0.0011**
		(0.0004)	(0.0005)		(0.0002)	(0.0003)		(0.0004)	(0.0004)
Δ_s Soil Moisture (m^3/m^3)		0.1934	0.1711		0.0887	0.1044		-0.0099	-0.0202
		(0.1663)	(0.1648)		(0.1001)	(0.0787)		(0.2264)	(0.2373)
Δ_s Slope Gradient (%)			-0.0259***			-0.0503***			0.0047
			(0.0060)			(0.0143)			(0.0069)
Δ_s Elevation (m)			0.0002**			0.0001			-0.0002***
			(7.06×10^{-5})			(9.77×10^{-5})			(6.44×10^{-5})
Δ_s Clay Content (%)			0.0035			0.0030**			0.0040*
			(0.0024)			(0.0012)			(0.0021)
R^2	0.00025	0.00093	0.00329	0.00374	0.00654	0.02017	0.00030	0.00312	0.00531
Observations	102,376	102,376	102,376	33,563	33,563	33,563	57,962	57,962	57,962
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The dependent variable is the spatial first difference of log crop yield. The treatment is the spatial difference in contemporaneous mean salinity. All differences are east minus west. Pairs link west-neighbor municipalities within the same State, Hydrological Basin, and latitude channel. Spec. 1 includes year fixed effects; Spec. 2 adds GDD, KDD, and soil moisture; Spec. 3 additionally controls for slope, elevation, and clay content. Salinity and agro-climatic controls are measured over the contemporaneous growing season. A crop-specific common complete-case sample is used across specifications. Conley (1999) spatial HAC standard errors (200 km cutoff; spherical distance) are in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

B.3 Spatial-inference robustness

The next four panels hold the point estimates fixed and vary only the cutoff used to compute the Conley spatial standard errors from 50 to 300 km. They report the complete excess-salinity and raw-mean-salinity results for all six crops.

Table 14: Conley Radius Robustness for Main Crop-Yield SFD Estimates: Excess Salinity: Panel A

Crop	Spec.	50 km	100 km	200 km	250 km	300 km	Obs.
		-0.085***	-0.085***	-0.085***	-0.085***	-0.085***	
Corn	Spec. 1	(0.021)	(0.021)	(0.025)	(0.023)	(0.024)	114,025
		-0.081***	-0.081***	-0.081***	-0.081***	-0.081***	
Corn	Spec. 2	(0.021)	(0.021)	(0.023)	(0.021)	(0.022)	114,025
		-0.073***	-0.073***	-0.073***	-0.073***	-0.073***	
Corn	Spec. 3	(0.020)	(0.020)	(0.022)	(0.020)	(0.021)	114,025
		0.042	0.042	0.042	0.042	0.042	
Rice	Spec. 1	(0.063)	(0.060)	(0.062)	(0.070)	(0.064)	66,072
		0.033	0.033	0.033	0.033	0.033	
Rice	Spec. 2	(0.059)	(0.054)	(0.054)	(0.065)	(0.055)	66,072
		0.012	0.012	0.012	0.012	$0.012^{\dagger}$	
Rice	Spec. 3	(0.052)	(0.047)	(0.044)	(0.054)	(0.038)	66,072
		-0.023	-0.023	-0.023	-0.023	-0.023	
Cassava	Spec. 1	(0.015)	(0.016)	(0.018)	(0.016)	(0.015)	94,296
		-0.023	-0.023	-0.023	-0.023	-0.023	
Cassava	Spec. 2	(0.015)	(0.016)	(0.017)	(0.016)	(0.015)	94,296
		-0.023	-0.023	-0.023	-0.023	-0.023	
Cassava	Spec. 3	(0.015)	(0.016)	(0.017)	(0.016)	(0.015)	94,296

Notes: Each cell reports the salinity coefficient and its Conley standard error. Coefficients and samples are identical across cutoffs within a row; only inference changes. Spec. 1 includes year fixed effects only, preferred Spec. 2 adds GDD, KDD, and soil moisture, and Spec. 3 adds slope, elevation, and clay content. Municipality pairs remain contiguous; the radius only governs residual spatial correlation. The tested radii are 50 km, 100 km, 200 km, 250 km, and 300 km. The 300 km column is a wider-radius sensitivity check. † indicates that the Conley variance-covariance matrix was not positive semi-definite and was adjusted by fixest. Significance levels: * p<0.10, ** p<0.05, *** p<0.01.

Table 15: Conley Radius Robustness for Main Crop-Yield SFD Estimates: Excess Salinity: Panel B

Crop	Spec.	50 km	100 km	200 km	250 km	300 km	Obs.
		-0.030*	-0.030*	-0.030*	-0.030**	-0.030**	
Beans	Spec. 1	(0.015)	(0.017)	(0.015)	(0.015)	(0.014)	102,376
		-0.029*	-0.029*	-0.029**	-0.029**	-0.029**	
Beans	Spec. 2	(0.016)	(0.018)	(0.015)	(0.014)	(0.012)	102,376
		-0.025*	-0.025	-0.025*	-0.025*	-0.025*	
Beans	Spec. 3	(0.015)	(0.017)	(0.014)	(0.014)	(0.013)	102,376
		-0.187	-0.187	-0.187	-0.187	-0.187***†	
Soybeans	Spec. 1	(0.254)	(0.281)	(0.249)	(0.148)	(<0.001)	33,563
		-0.135	-0.135	-0.135	-0.135†	-0.135***†	
Soybeans	Spec. 2	(0.253)	(0.283)	(0.224)	(0.083)	(0.010)	33,563
		-0.167	-0.167	-0.167†	-0.167†	-0.167***†	
Soybeans	Spec. 3	(0.231)	(0.255)	(0.230)	(0.126)	(0.039)	33,563
		-0.009	-0.009	-0.009	-0.009	-0.009	
Sugarcane	Spec. 1	(0.033)	(0.033)	(0.012)	(0.028)	(0.023)	57,069
		-0.015	-0.015	-0.015	-0.015	-0.015	
Sugarcane	Spec. 2	(0.033)	(0.031)	(0.012)	(0.028)	(0.022)	57,069
		-0.022	-0.022	-0.022	-0.022†	-0.022†	
Sugarcane	Spec. 3	(0.031)	(0.031)	(0.015)	(0.027)	(0.022)	57,069

Notes: Each cell reports the salinity coefficient and its Conley standard error. Coefficients and samples are identical across cutoffs within a row; only inference changes. Spec. 1 includes year fixed effects only, preferred Spec. 2 adds GDD, KDD, and soil moisture, and Spec. 3 adds slope, elevation, and clay content. Municipality pairs remain contiguous; the radius only governs residual spatial correlation. The tested radii are 50 km, 100 km, 200 km, 250 km, and 300 km. The 300 km column is a wider-radius sensitivity check. † indicates that the Conley variance-covariance matrix was not positive semi-definite and was adjusted by fixest. Significance levels: * p<0.10, ** p<0.05, *** p<0.01.

Table 16: Conley Radius Robustness for Main Crop-Yield SFD Estimates: Mean Salinity: Panel A

Crop	Spec.	50 km	100 km	200 km	250 km	300 km	Obs.
		-0.061***	-0.061***	-0.061***	-0.061***	-0.061***	
Corn	Spec. 1	(0.014)	(0.014)	(0.015)	(0.014)	(0.015)	114,025
		-0.060***	-0.060***	-0.060***	-0.060***	-0.060***	
Corn	Spec. 2	(0.014)	(0.014)	(0.014)	(0.013)	(0.014)	114,025
		-0.053***	-0.053***	-0.053***	-0.053***	-0.053***	
Corn	Spec. 3	(0.014)	(0.013)	(0.013)	(0.011)	(0.013)	114,025
		0.020	0.020	0.020	0.020	0.020	
Rice	Spec. 1	(0.034)	(0.031)	(0.032)	(0.034)	(0.038)	66,072
		0.016	0.016	0.016	0.016	0.016	
Rice	Spec. 2	(0.032)	(0.028)	(0.030)	(0.032)	(0.034)	66,072
		0.015	0.015	0.015	0.015	0.015†	
Rice	Spec. 3	(0.029)	(0.025)	(0.024)	(0.027)	(0.028)	66,072
		-0.014	-0.014	-0.014	-0.014	-0.014	
Cassava	Spec. 1	(0.011)	(0.013)	(0.013)	(0.012)	(0.012)	94,296
		-0.014	-0.014	-0.014	-0.014	-0.014	
Cassava	Spec. 2	(0.012)	(0.013)	(0.013)	(0.012)	(0.012)	94,296
		-0.013	-0.013	-0.013	-0.013	-0.013	
Cassava	Spec. 3	(0.012)	(0.013)	(0.013)	(0.012)	(0.012)	94,296

Notes: Each cell reports the salinity coefficient and its Conley standard error. Coefficients and samples are identical across cutoffs within a row; only inference changes. Spec. 1 includes year fixed effects only, preferred Spec. 2 adds GDD, KDD, and soil moisture, and Spec. 3 adds slope, elevation, and clay content. Municipality pairs remain contiguous; the radius only governs residual spatial correlation. The tested radii are 50 km, 100 km, 200 km, 250 km, and 300 km. The 300 km column is a wider-radius sensitivity check. † indicates that the Conley variance-covariance matrix was not positive semi-definite and was adjusted by fixest. Significance levels: * p<0.10, ** p<0.05, *** p<0.01.

Table 17: Conley Radius Robustness for Main Crop-Yield SFD Estimates: Mean Salinity: Panel B

Crop	Spec.	50 km	100 km	200 km	250 km	300 km	Obs.
		-0.024*	-0.024	-0.024*	-0.024*	-0.024*	
Beans	Spec. 1	(0.013)	(0.015)	(0.014)	(0.014)	(0.013)	102,376
		-0.025*	-0.025	-0.025*	-0.025*	-0.025**	
Beans	Spec. 2	(0.014)	(0.015)	(0.013)	(0.013)	(0.011)	102,376
		-0.020	-0.020	-0.020	-0.020	-0.020*	
Beans	Spec. 3	(0.013)	(0.014)	(0.012)	(0.012)	(0.012)	102,376
		-0.070***	-0.070***	-0.070***	-0.070***	-0.070***	
Soybeans	Spec. 1	(0.018)	(0.021)	(0.018)	(0.016)	(0.010)	33,563
		-0.065***	-0.065***	-0.065***	-0.065***†	-0.065***†	
Soybeans	Spec. 2	(0.018)	(0.020)	(0.017)	(0.013)	(0.002)	33,563
		-0.042***	-0.042**	-0.042**†	-0.042**†	-0.042***†	
Soybeans	Spec. 3	(0.016)	(0.019)	(0.018)	(0.017)	(0.010)	33,563
		-0.004	-0.004	-0.004	-0.004	-0.004	
Sugarcane	Spec. 1	(0.022)	(0.022)	(0.010)	(0.016)	(0.013)	57,962
		-0.007	-0.007	-0.007	-0.007	-0.007	
Sugarcane	Spec. 2	(0.021)	(0.020)	(0.009)	(0.016)	(0.012)	57,962
		-0.008	-0.008	-0.008	-0.008†	-0.008†	
Sugarcane	Spec. 3	(0.020)	(0.019)	(0.010)	(0.016)	(0.011)	57,962

Notes: Each cell reports the salinity coefficient and its Conley standard error. Coefficients and samples are identical across cutoffs within a row; only inference changes. Spec. 1 includes year fixed effects only, preferred Spec. 2 adds GDD, KDD, and soil moisture, and Spec. 3 adds slope, elevation, and clay content. Municipality pairs remain contiguous; the radius only governs residual spatial correlation. The tested radii are 50 km, 100 km, 200 km, 250 km, and 300 km. The 300 km column is a wider-radius sensitivity check. † indicates that the Conley variance-covariance matrix was not positive semi-definite and was adjusted by fixest. Significance levels: * p<0.10, ** p<0.05, *** p<0.01.

B.4 North-south orientation

The north-south construction retains strict contiguity and the state and hydrological-basin restrictions. The negative corn coefficient is reproduced with both excess and raw mean salinity. Rice, cassava, and sugarcane remain statistically insignificant. The bean coefficient remains negative but is less precise. Several soybean covariance matrices require a numerical positive-semidefinite adjustment, so this orientation does not provide an independent validation of soybean precision.

Table 18: North–South Yield Robustness: Excess Salinity: Panel A

Crop	Corn			Rice			Cassava		
Spec	Spec. 1	Spec. 2	Spec. 3	Spec. 1	Spec. 2	Spec. 3	Spec. 1	Spec. 2	Spec. 3
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Variables									
Δ_{N-S} Excess salinity (dS/m)	-0.0486**	-0.0493**	-0.0395*	0.0148	0.0171	0.0022	-0.0022	-0.0022	-0.0017
	(0.0225)	(0.0219)	(0.0204)	(0.0377)	(0.0381)	(0.0334)	(0.0122)	(0.0122)	(0.0123)
Δ_{N-S} GDD		-9.26×10^{-6}	5.1×10^{-5}**		9.02×10^{-5}*	-1.54×10^{-5}		-2.01×10^{-6}	-1.84×10^{-6}
		(3.3×10^{-5})	(2.39×10^{-5})		(5.37×10^{-5})	(4.02×10^{-5})		(9.42×10^{-6})	(9.03×10^{-6})
Δ_{N-S} KDD		-0.0008***	-0.0005		-0.0012	-0.0035		-7.29×10^{-5}	-5.2×10^{-5}
		(0.0002)	(0.0003)		(0.0049)	(0.0054)		(0.0001)	(0.0001)
Δ_{N-S} Soil moisture		0.2248	0.1690		0.4506*	0.4082*		0.0220	0.0179
		(0.2216)	(0.2141)		(0.2403)	(0.2199)		(0.1398)	(0.1445)
Δ_{N-S} Slope			-0.0258***			-0.0105			-0.0015
			(0.0079)			(0.0138)			(0.0025)
Δ_{N-S} Elevation			0.0002**			-0.0003**			1.28×10^{-5}
			(7.68×10^{-5})			(0.0001)			(4.3×10^{-5})
Δ_{N-S} Clay			0.0084***			0.0071**			0.0004
			(0.0021)			(0.0028)			(0.0015)
Fixed-effects									
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics									
R^2	0.00044	0.00224	0.00643	0.00021	0.00178	0.00496	0.00011	0.00019	0.00022
Observations	123,689	123,689	123,689	72,833	72,833	72,833	102,675	102,675	102,675

Conley (200km) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: The dependent variable is the north-minus-south difference in log crop yield. Pairs are strictly contiguous municipalities in the same State, hydrological basin and longitude channel; the nearest eligible southern neighbour is retained. The treatment is crop-specific excess salinity above the FAO threshold. Spec. 1 includes year fixed effects; preferred Spec. 2 adds GDD, KDD and soil moisture; Spec. 3 adds slope, elevation and clay. Each crop uses one common complete-data sample across specifications. Conley spatial standard errors use a 200 km cutoff. The following covariance matrices required a positive semi-definite adjustment: Soybeans - Spec. 1, Soybeans - Spec. 2, Soybeans - Spec. 3.

Table 19: North–South Yield Robustness: Excess Salinity: Panel B

Crop	Beans			Soybeans			Sugarcane		
Spec	Spec. 1	Spec. 2	Spec. 3	Spec. 1	Spec. 2	Spec. 3	Spec. 1	Spec. 2	Spec. 3
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Variables									
Δ_{N-S} Excess salinity (dS/m)	-0.0103	-0.0102	-0.0035	-0.1639***	-0.1210***	-0.1626***	-0.0054	-0.0097	-0.0190
	(0.0127)	(0.0131)	(0.0131)	(1×10^{-6})	(0.0213)	(0.0260)	(0.0524)	(0.0564)	(0.0527)
Δ_{N-S} GDD		-6.89×10^{-5}	-7.89×10^{-6}		-7.97×10^{-5}	8.06×10^{-5}		0.0001*	4.37×10^{-5}
		(4.82×10^{-5})	(2.7×10^{-5})		(0.0001)	(6.5×10^{-5})		(8.92×10^{-5})	(8.17×10^{-5})
Δ_{N-S} KDD		-0.0002	8.88×10^{-5}		-0.0005	-0.0006		-0.0008	-0.0014**
		(0.0003)	(0.0003)		(0.0004)	(0.0004)		(0.0005)	(0.0006)
Δ_{N-S} Soil moisture		0.0353	2.43×10^{-5}		0.1611	0.1369		-0.0123	-0.0200
		(0.1952)	(0.2062)		(0.1125)	(0.1060)		(0.1607)	(0.1602)
Δ_{N-S} Slope			-0.0286***			-0.0480**			0.0055
			(0.0059)			(0.0215)			(0.0077)
Δ_{N-S} Elevation			0.0002**			0.0002			-0.0003***
			(7.58×10^{-5})			(0.0002)			(9.29×10^{-5})
Δ_{N-S} Clay			0.0043*			0.0026			0.0042
			(0.0023)			(0.0019)			(0.0026)
Fixed-effects									
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics									
R^2	0.00021	0.00083	0.00372	0.00068	0.00456	0.01570	0.00021	0.00219	0.00536
Observations	110,506	110,506	110,506	36,958	36,958	36,958	60,979	60,979	60,979

Conley (200km) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: The dependent variable is the north-minus-south difference in log crop yield. Pairs are strictly contiguous municipalities in the same State, hydrological basin and longitude channel; the nearest eligible southern neighbour is retained. The treatment is crop-specific excess salinity above the FAO threshold. Spec. 1 includes year fixed effects; preferred Spec. 2 adds GDD, KDD and soil moisture; Spec. 3 adds slope, elevation and clay. Each crop uses one common complete-data sample across specifications. Conley spatial standard errors use a 200 km cutoff. The following covariance matrices required a positive semi-definite adjustment: Soybeans - Spec. 1, Soybeans - Spec. 2, Soybeans - Spec. 3.

Table 20: North–South Yield Robustness: Raw Mean Salinity: Panel A

Crop	Corn			Rice			Cassava		
Spec	Spec. 1	Spec. 2	Spec. 3	Spec. 1	Spec. 2	Spec. 3	Spec. 1	Spec. 2	Spec. 3
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Variables									
Δ_{N-S} Raw mean salinity (dS/m)	-0.0391**	-0.0414***	-0.0329**	0.0231	0.0235	0.0210	0.0004	0.0002	0.0006
	(0.0162)	(0.0157)	(0.0142)	(0.0225)	(0.0230)	(0.0191)	(0.0100)	(0.0100)	(0.0101)
Δ_{N-S} GDD		-8.82×10^{-6}	5.14×10^{-5}**		9.05×10^{-5}*	-1.21×10^{-5}		-2.02×10^{-6}	-1.84×10^{-6}
		(3.29×10^{-5})	(2.38×10^{-5})		(5.33×10^{-5})	(3.93×10^{-5})		(9.42×10^{-6})	(9.04×10^{-6})
Δ_{N-S} KDD		-0.0008***	-0.0005*		-0.0012	-0.0034		-7.28×10^{-5}	-5.07×10^{-5}
		(0.0002)	(0.0003)		(0.0049)	(0.0053)		(0.0001)	(0.0001)
Δ_{N-S} Soil moisture		0.2284	0.1724		0.4466*	0.4034*		0.0220	0.0176
		(0.2208)	(0.2136)		(0.2420)	(0.2207)		(0.1399)	(0.1448)
Δ_{N-S} Slope			-0.0256***			-0.0103			-0.0015
			(0.0078)			(0.0138)			(0.0025)
Δ_{N-S} Elevation			0.0002**			-0.0003**			1.33×10^{-5}
			(7.66×10^{-5})			(0.0001)			(4.28×10^{-5})
Δ_{N-S} Clay			0.0084***			0.0073**			0.0004
			(0.0021)			(0.0028)			(0.0015)
Fixed-effects									
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics									
R^2	0.00054	0.00239	0.00651	0.00036	0.00193	0.00510	0.00011	0.00019	0.00022
Observations	123,689	123,689	123,689	72,833	72,833	72,833	102,675	102,675	102,675

Conley (200km) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: The dependent variable is the north-minus-south difference in log crop yield. Pairs are strictly contiguous municipalities in the same State, hydrological basin and longitude channel; the nearest eligible southern neighbour is retained. The treatment is raw mean soil salinity. Spec. 1 includes year fixed effects; preferred Spec. 2 adds GDD, KDD and soil moisture; Spec. 3 adds slope, elevation and clay. Each crop uses one common complete-data sample across specifications. Conley spatial standard errors use a 200 km cutoff. The following covariance matrices required a positive semi-definite adjustment: Soybeans - Spec. 2, Soybeans - Spec. 3.

Table 21: North–South Yield Robustness: Raw Mean Salinity: Panel B

Crop	Beans			Soybeans			Sugarcane		
Spec	Spec. 1	Spec. 2	Spec. 3	Spec. 1	Spec. 2	Spec. 3	Spec. 1	Spec. 2	Spec. 3
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Variables									
Δ_{N-S} Raw mean salinity (dS/m)	-0.0054	-0.0061	-0.0008	-0.0387***	-0.0352***	-0.0189**	-0.0040	-0.0078	-0.0120
	(0.0113)	(0.0113)	(0.0114)	(0.0081)	(0.0040)	(0.0081)	(0.0327)	(0.0344)	(0.0330)
Δ_{N-S} GDD		-6.91×10^{-5}	-7.65×10^{-6}		-7.56×10^{-5}	7.79×10^{-5}		0.0001*	4.56×10^{-5}
		(4.82×10^{-5})	(2.7×10^{-5})		(0.0001)	(0.0001)		(8.81×10^{-5})	(8.05×10^{-5})
Δ_{N-S} KDD		-0.0002	9.12×10^{-5}		-0.0005	-0.0006		-0.0008	-0.0014**
		(0.0003)	(0.0003)		(0.0003)	(0.0005)		(0.0005)	(0.0006)
Δ_{N-S} Soil moisture		0.0352	-0.0004		0.1800	0.1456		-0.0188	-0.0270
		(0.1952)	(0.2064)		(0.1103)	(0.1074)		(0.1592)	(0.1589)
Δ_{N-S} Slope			-0.0285***			-0.0468***			0.0060
			(0.0059)			(0.0173)			(0.0078)
Δ_{N-S} Elevation			0.0002**			0.0002			-0.0003***
			(7.6×10^{-5})			(0.0003)			(9.38×10^{-5})
Δ_{N-S} Clay			0.0044*			0.0026			0.0043
			(0.0023)			(0.0017)			(0.0026)
Fixed-effects									
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics									
R^2	0.00019	0.00082	0.00372	0.00166	0.00539	0.01582	0.00022	0.00222	0.00532
Observations	110,506	110,506	110,506	36,958	36,958	36,958	61,962	61,962	61,962

Conley (200km) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: The dependent variable is the north-minus-south difference in log crop yield. Pairs are strictly contiguous municipalities in the same State, hydrological basin and longitude channel; the nearest eligible southern neighbour is retained. The treatment is raw mean soil salinity. Spec. 1 includes year fixed effects; preferred Spec. 2 adds GDD, KDD and soil moisture; Spec. 3 adds slope, elevation and clay. Each crop uses one common complete-data sample across specifications. Conley spatial standard errors use a 200 km cutoff. The following covariance matrices required a positive semi-definite adjustment: Soybeans - Spec. 2, Soybeans - Spec. 3.

C Crop Output

C.1 Detailed estimates and functional-form comparison

Table 22 reports the raw coefficients, crop-specific treatment standard deviations, standardized effects, sample sizes, and controls underlying the output figure. Table 23 compares the log specification with the asinh specifications on the zero-inclusive and common positive-output samples.

Table 22: Crop Output Responses to Excess Soil Salinity

Crop	Specification	Raw coefficient	Treatment SD	One-SD effect	One-SD SE	Observations	Pairs	GDD	KDD	Soil moisture	Elevation	Slope	Clay	PSD adjustment
Corn	Spec. 1	-0.2360**	0.1642	-0.0388**	0.0170	114,025	4,070	No	No	No	No	No	No	No
Corn	Spec. 2	-0.2149**	0.1642	-0.0353**	0.0146	114,025	4,070	Yes	Yes	Yes	No	No	No	No
Corn	Spec. 3	-0.1740**	0.1642	-0.0286**	0.0145	114,025	4,070	Yes	Yes	Yes	Yes	Yes	Yes	No
Rice	Spec. 1	0.0913	0.1118	0.0102	0.0111	66,072	3,280	No	No	No	No	No	No	No
Rice	Spec. 2	0.0492	0.1118	0.0055	0.0119	66,072	3,280	Yes	Yes	Yes	No	No	No	No
Rice	Spec. 3	-0.1117**	0.1118	-0.0125**	0.0062	66,072	3,280	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cassava	Spec. 1	0.1397*	0.1893	0.0264*	0.0158	94,296	3,860	No	No	No	No	No	No	No
Cassava	Spec. 2	0.1558*	0.1893	0.0295*	0.0155	94,296	3,860	Yes	Yes	Yes	No	No	No	No
Cassava	Spec. 3	0.1077*	0.1893	0.0204*	0.0123	94,296	3,860	Yes	Yes	Yes	Yes	Yes	Yes	No
Beans	Spec. 1	-0.0069	0.2105	-0.0015	0.0246	102,376	4,027	No	No	No	No	No	No	No
Beans	Spec. 2	-0.0008	0.2105	-0.0002	0.0242	102,376	4,027	Yes	Yes	Yes	No	No	No	No
Beans	Spec. 3	0.0395	0.2105	0.0083	0.0245	102,376	4,027	Yes	Yes	Yes	Yes	Yes	Yes	No
Soybeans	Spec. 1	-5.0806***	0.0153	-0.0776***	0.0000	33,563	1,774	No	No	No	No	No	No	Yes
Soybeans	Spec. 2	-4.1525***	0.0153	-0.0634***	0.0014	33,563	1,774	Yes	Yes	Yes	No	No	No	Yes
Soybeans	Spec. 3	-5.0258***	0.0153	-0.0768***	0.0009	33,563	1,774	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sugarcane	Spec. 1	0.1278	0.1420	0.0181	0.0136	57,069	3,117	No	No	No	No	No	No	No
Sugarcane	Spec. 2	0.0676	0.1420	0.0096	0.0095	57,069	3,117	Yes	Yes	Yes	No	No	No	No
Sugarcane	Spec. 3	0.0369	0.1420	0.0052	0.0122	57,069	3,117	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The dependent variable is the east-minus-west difference in log physical output quantity. The treatment is the east-minus-west difference in crop-specific excess salinity in dS/m. One-SD effects multiply the raw coefficient and its standard error by the crop-specific standard deviation of the treatment; this rescaling does not change p-values or statistical significance. All specifications include year fixed effects. Spec. 2 is preferred and adds GDD, KDD and soil moisture; Spec. 3 also adds slope, elevation and clay. Each crop uses one common complete-data sample across specifications. Standard errors are Conley spatial with a 200 km cutoff.

Table 23: Crop Output: Log and Asinh Specifications

Crop	Outcome	Raw coefficient	Standard error	Treatment SD	One-SD effect	Observations	Pairs
Corn	Log, positive-output sample	-0.2149**	(0.0890)	0.1642	-0.0353	114,025	4,070
Corn	Asinh, same positive-output sample	-0.2148**	(0.0889)	0.1642	-0.0353	114,025	4,070
Corn	Asinh, short zero spells retained	-0.2148**	(0.0889)	0.1642	-0.0353	114,034	4,070
Rice	Log, positive-output sample	0.0492	(0.1064)	0.1118	0.0055	66,072	3,280
Rice	Asinh, same positive-output sample	0.0489	(0.1069)	0.1118	0.0055	66,072	3,280
Rice	Asinh, short zero spells retained	0.0489	(0.1069)	0.1118	0.0055	66,072	3,280
Cassava	Log, positive-output sample	0.1558*	(0.0818)	0.1893	0.0295	94,296	3,860
Cassava	Asinh, same positive-output sample	0.1558*	(0.0818)	0.1893	0.0295	94,296	3,860
Cassava	Asinh, short zero spells retained	0.1558*	(0.0818)	0.1893	0.0295	94,297	3,860
Beans	Log, positive-output sample	-0.0008	(0.1151)	0.2105	-0.0002	102,376	4,027
Beans	Asinh, same positive-output sample	0.0006	(0.1152)	0.2105	0.0001	102,376	4,027
Beans	Asinh, short zero spells retained	0.0006	(0.1152)	0.2105	0.0001	102,386	4,027
Soybeans	Log, positive-output sample	-4.1525***	(0.0900)	0.0153	-0.0634	33,563	1,774
Soybeans	Asinh, same positive-output sample	-4.1472***	(0.0895)	0.0153	-0.0633	33,563	1,774
Soybeans	Asinh, short zero spells retained	-4.1472***	(0.0895)	0.0153	-0.0633	33,563	1,774
Sugarcane	Log, positive-output sample	0.0676	(0.0667)	0.1420	0.0096	57,069	3,117
Sugarcane	Asinh, same positive-output sample	0.0676	(0.0667)	0.1420	0.0096	57,069	3,117
Sugarcane	Asinh, short zero spells retained	0.0457	(0.1284)	0.1418	0.0065	60,529	3,146

Notes: All rows report preferred Spec. 2 with year fixed effects, GDD, KDD and soil moisture, and Conley spatial standard errors with a 200 km cutoff. The log and same-sample asinh rows use exactly the same crop-specific complete-data sample with positive output on both sides of the spatial pair. Their difference isolates the functional form. The zero-inclusive asinh row additionally retains zero output when it belongs to a spell shorter than five consecutive years; longer zero runs are treated as structural missing values. Differences between the second and third rows therefore also reflect the enlarged sample and transitions into or out of zero output. Asinh coefficients approximate log changes for large positive quantities but do not have a percentage interpretation for transitions involving zero.

C.2 Alternative clustered inference

Table 24: Crop Output: Conley and Pair-Clustered Inference

Crop	Raw coefficient	Conley SE (200 km)	Conley p-value	Pair-clustered SE	Pair-clustered p-value	Observations	Pairs
Corn	-0.2149	0.0890	0.016	0.0854	0.012	114,025	4,070
Rice	0.0492	0.1064	0.644	0.2385	0.837	66,072	3,280
Cassava	0.1558	0.0818	0.057	0.0820	0.057	94,296	3,860
Beans	-0.0008	0.1151	0.995	0.0837	0.993	102,376	4,027
Soybeans	-4.1525	0.0900	0.000	2.4683	0.093	33,563	1,774
Sugarcane	0.0676	0.0667	0.310	0.1616	0.676	57,069	3,117

Notes: The preferred crop-output specification, coefficient, and estimation sample are held fixed. Only the covariance estimator changes: Conley spatial with a 200 km cutoff versus clustering by spatial municipality pair. Pair clustering allows unrestricted dependence over time within an exact pair but does not replace the main Conley correction for dependence across nearby pairs.

C.3 Spatial-inference robustness

Table 25 retains the preferred specification and varies only the cutoff used to compute the Conley spatial standard errors. An `NA` entry indicates that the strict Conley covariance matrix is not positive semidefinite and therefore does not provide a valid estimate of precision.

Table 25: Spatial-Inference Robustness for Next-Season Output

Crop	Pair cluster	50 km	100 km	200 km	250 km	300 km
	-0.0023	-0.0023	-0.0023	-0.0023	-0.0023	-0.0023
Corn	(0.0018)	(0.0019)	(0.0019)	(0.0021)	(0.0022)	(0.0025)
	0.0009	0.0009	0.0009	0.0009	0.0009	0.0009
Rice	(0.0024)	(0.0024)	(0.0023)	(0.0006)	(0.0015)	(0.0017)
	0.0012	0.0012	0.0012	0.0012	0.0012	0.0012
Cassava	(0.0015)	(0.0014)	(0.0015)	(0.0015)	(0.0015)	(0.0015)
	-0.0026	-0.0026	-0.0026	-0.0026	-0.0026	-0.0026
Beans	(0.0019)	(0.0021)	(0.0023)	(0.0018)	(0.0019)	(0.0016)
	0.0064	0.0064*	0.0064	0.0064	0.0064**	
Soybeans	(0.0040)	(0.0039)	(0.0062)	(0.0050)	(0.0026)	NA
	-0.0053	-0.0053	-0.0053	-0.0053	-0.0053	-0.0053
Sugarcane	(0.0065)	(0.0084)	(0.0060)	(0.0053)	(0.0055)	(0.0053)

Notes: Cells report the preferred-specification effect of a one-standard-deviation increase in crop-specific excess salinity, with standard errors in parentheses. All regressions include year fixed effects and changes from t to t+1 in GDD, KDD and soil moisture. Point estimates and samples are identical across columns; only the variance estimator changes. Strict Conley estimates use spherical distance and no positive-semidefinite covariance correction. NA indicates that a valid strict covariance estimate could not be obtained.

C.4 Soybean treatment-support diagnostic

The soybean diagnostic compares continuous excess salinity in the full sample, the sample restricted to exposed pairs, threshold crossing, and raw mean salinity. Raw mean salinity defines a different estimand and is included only to document treatment support.

Table 26: Soybean Excess-Salinity Support Diagnostics

Outcome	Exposure	Coefficient	Std. error	P-value	Observations	Pairs
Planted area	Crop-specific excess: full sample	0.4648	(0.6532)	0.477	28,302	1,609
Planted area	Crop-specific excess: exposed pairs	0.5000	(0.6647)	0.452	2,704	333
Planted area	Difference in threshold exceedance	0.0302	(0.0267)	0.258	28,302	1,609
Planted area	Asinh crop-specific excess	0.4736	(0.6818)	0.487	28,302	1,609
Planted area	Continuous mean salinity (different estimand)	0.0280	(0.0314)	0.373	28,302	1,609
Production quantity	Crop-specific excess: full sample	0.4408	(0.7150)	0.538	28,302	1,609
Production quantity	Crop-specific excess: exposed pairs	0.4728	(0.7313)	0.518	2,704	333
Production quantity	Difference in threshold exceedance	0.0467	(0.0305)	0.126	28,302	1,609
Production quantity	Asinh crop-specific excess	0.4377	(0.7485)	0.559	28,302	1,609
Production quantity	Continuous mean salinity (different estimand)	0.0256	(0.0347)	0.460	28,302	1,609

Notes: All regressions use the main soybean sample, year fixed effects, crop-season GDD, KDD and soil-moisture controls, and standard errors clustered by spatial pair. The exposed-pair regression retains only pair-years in which at least one municipality has positive excess above the 5 dS/m threshold. The threshold-exceedance coefficient compares a one-unit east-minus-west difference in exposure status. Continuous mean salinity uses dS/m but estimates a different response from excess above the soybean threshold.

D Crop Producer Prices

The following panels report the complete municipal producer-price regressions for the six crops. The outcome is the east-minus-west difference in log real unit value, and the treatment is crop-specific excess salinity.

Table 27: Soil Salinity and Municipal Producer Prices: Panel A

Crop		Corn			Rice			Cassava	
Spec	Spec. 1	Spec. 2	Spec. 3	Spec. 1	Spec. 2	Spec. 3	Spec. 1	Spec. 2	Spec. 3
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Variables									
Δ_s Excess salinity (dS/m)	0.0017	-0.0021	-0.0026	0.0092	-0.0109	-0.0118	-0.0174	-0.0144	-0.0159*
	(0.0059)	(0.0091)	(0.0086)	(0.0109)	(0.0151)	(0.0158)	(0.0120)	(0.0090)	(0.0090)
Δ_s GDD	-1.43×10^{-5}	-4.16×10^{-5}	-4.44×10^{-5}	-8.79×10^{-6}	-0.0002**	-0.0002**	4.29×10^{-6}	3.42×10^{-5}	3.28×10^{-5}
	(9.02×10^{-6})	(5.45×10^{-5})	(5.54×10^{-5})	(1.28×10^{-5})	(7.6×10^{-5})	(7.55×10^{-5})	(7.9×10^{-6})	(5.11×10^{-5})	(5.21×10^{-5})
Δ_s KDD	0.0002***	-0.0002	-0.0002	-0.0004	-0.0004	-0.0002	0.0002	0.0002	0.0002
	(5.86×10^{-5})	(0.0001)	(0.0001)	(0.0007)	(0.0005)	(0.0004)	(0.0002)	(0.0002)	(0.0003)
Δ_s Soil moisture	-0.0630	0.1613	0.1634	-0.0170	-0.1887	-0.2064	-0.0062	0.0777	0.0703
	(0.0416)	(0.1294)	(0.1328)	(0.0547)	(0.1875)	(0.1869)	(0.1276)	(0.1706)	(0.1731)
Δ_s Elevation	1.27×10^{-5}			1.28×10^{-6}			-2.78×10^{-5}		
	(1.65×10^{-5})			(3.2×10^{-5})			(3.98×10^{-5})		
Δ_s Slope	0.0004			-0.0027			0.0021		
	(0.0013)			(0.0022)			(0.0052)		
Δ_s Clay	1.62×10^{-5}			-0.0004			0.0040***		
	(0.0004)			(0.0008)			(0.0012)		
Fixed-effects									
Year	Yes	Yes		Yes	Yes		Yes	Yes	
pair_id		Yes	Yes		Yes	Yes		Yes	Yes
state-Year			Yes			Yes			Yes
Fit statistics									
R^2	0.00036	0.09058	0.09464	0.00028	0.11541	0.12127	0.00113	0.15417	0.15845
Observations	113,728	113,724	113,641	65,664	65,632	65,558	94,202	94,136	94,050

Conley (200km) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: The dependent variable is the east-minus-west difference in the log PAM average producer price, equal to production value divided by production quantity. Spec. 1 includes year fixed effects, weather, soil and topographic controls. Spec. 2 includes pair and year fixed effects plus weather controls; Spec. 3 replaces year fixed effects with state-by-year fixed effects and is preferred. Time-invariant soil and topographic differences are absorbed by pair fixed effects in Specs. 2–3. All specifications start from a crop-specific common complete-data sample; observation counts can differ because singleton fixed-effect groups are removed in Specs. 2–3. Standard errors are Conley spatial with a 200 km cutoff.

Table 28: Soil Salinity and Municipal Producer Prices: Panel B

Crop		Beans			Soybeans			Sugarcane	
Spec	Spec. 1	Spec. 2	Spec. 3	Spec. 1	Spec. 2	Spec. 3	Spec. 1	Spec. 2	Spec. 3
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Variables									
Δ_s Excess salinity (dS/m)	-0.0064	-0.0081	-0.0082	0.0567	0.1937*	0.1922*	0.0644**	0.0381	0.0401
	(0.0069)	(0.0102)	(0.0098)	(0.0683)	(0.1142)	(0.1111)	(0.0271)	(0.0246)	(0.0249)
Δ_s GDD	1.16×10^{-5}	-1.38×10^{-5}	-1.18×10^{-5}	5.66×10^{-5}	-0.0001*	-0.0001*	-1.15×10^{-5}	-1.36×10^{-6}	-5.44×10^{-6}
	(1.82×10^{-5})	(5.13×10^{-5})	(5.26×10^{-5})	(4.01×10^{-5})	(7.42×10^{-5})	(7.62×10^{-5})	(2.88×10^{-5})	(6.61×10^{-5})	(6.85×10^{-5})
Δ_s KDD	-9.14×10^{-5}	-0.0002	-0.0002	-0.0003**	0.0002	0.0002	0.0004	0.0006**	0.0005*
	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0003)	(0.0003)	(0.0004)	(0.0003)	(0.0003)
Δ_s Soil moisture	-0.0087	0.3145***	0.3151***	0.0222	0.0056	0.0044	-0.1360	-0.0561	-0.0651
	(0.0640)	(0.0738)	(0.0741)	(0.0311)	(0.0979)	(0.0976)	(0.1358)	(0.2497)	(0.2479)
Δ_s Elevation	2.05×10^{-5}			2.98×10^{-5}			3.71×10^{-5}		
	(2.95×10^{-5})			(4.12×10^{-5})			(7.08×10^{-5})		
Δ_s Slope	0.0022			-0.0034*			-0.0032		
	(0.0025)			(0.0020)			(0.0060)		
Δ_s Clay	-0.0001			0.0003			-0.0025		
	(0.0007)			(0.0003)			(0.0020)		
Fixed-effects									
Year	Yes	Yes		Yes	Yes		Yes	Yes	
pair_id		Yes	Yes		Yes	Yes		Yes	Yes
state-Year			Yes			Yes			Yes
Fit statistics									
R^2	0.00036	0.09936	0.10309	0.00125	0.09426	0.10101	0.00173	0.17104	0.17644
Observations	102,305	102,271	102,199	33,535	33,448	33,401	57,828	57,688	57,635

Conley (200km) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: The dependent variable is the east-minus-west difference in the log PAM average producer price, equal to production value divided by production quantity. Spec. 1 includes year fixed effects, weather, soil and topographic controls. Spec. 2 includes pair and year fixed effects plus weather controls; Spec. 3 replaces year fixed effects with state-by-year fixed effects and is preferred. Time-invariant soil and topographic differences are absorbed by pair fixed effects in Specs. 2–3. All specifications start from a crop-specific common complete-data sample; observation counts can differ because singleton fixed-effect groups are removed in Specs. 2–3. Standard errors are Conley spatial with a 200 km cutoff.

E Historical Exposure and Yield Sensitivity

Table 29 reports the two-step Butler and Huybers (2013) diagnostic discussed in the main text. The first step estimates municipality-specific yield sensitivity, and the second relates these estimates to historical salinity exposure.

Table 29: Butler–Huybers Historical-Adaptation Diagnostic

Crop	Municipalities	Factor (bootstrap SE)	95% CI	p-value	Weighted R2
Corn	3,885	0.023** (0.009)	[0.007, 0.044]	0.010	0.004
Rice	2,782	0.055** (0.019)	[0.019, 0.094]	0.012	0.004
Cassava	3,521	0.005*** (0.007)	[0.002, 0.026]	0.004	0.001
Beans	3,696	-0.005 (0.017)	[-0.027, 0.039]	0.964	0.000
Soybeans	322	-0.012 (0.322)	[-0.434, 0.844]	0.972	0.000
Sugarcane	2,447	0.044*** (0.017)	[0.012, 0.080]	0.008	0.008

Notes: Stage 1 estimates a separate municipal crop-yield sensitivity to annual crop-specific excess-salinity anomalies over 1985–2018, controlling for a municipality-specific linear trend and within-municipality GDD, KDD and soil-moisture anomalies. Municipality–crop series require at least 15 positive-yield years and at least five salinity observations in 1985–1994. Stage-1 uncertainty uses 199 circular moving-block residual bootstrap replications with three-year blocks. Following Butler and Huybers (2013), Stage 2 regresses the estimated sensitivities on log historical mean salinity, weighting by the inverse stage-1 bootstrap variance. A positive adaptation factor means that sensitivity becomes less negative as historical salinity rises and is therefore compatible with adaptation; it is not a causal estimate of adaptation. Stage-2 confidence intervals and p-values use 999 municipality bootstrap replications.

F Crop-Specific Planted Area

Table 30 reports the three specifications underlying the crop-specific planted-area figure. An observation is a spatial municipality pair, a crop, and a baseline campaign between 1988 and 2017. Planted area is positive at baseline in both municipalities, while zero planted area in the following campaign is retained.

Table 30: Excess Salinity and Change in Planted Area

Panel A: Corn, Rice and Cassava

Crop	Corn			Rice			Cassava		
Specification	Spec. 1	Spec. 2	Spec. 3	Spec. 1	Spec. 2	Spec. 3	Spec. 1	Spec. 2	Spec. 3
Δ_s Excess salinity at t (dS/m)	-0.0399***	-0.0391***	-0.0379***	0.0044	0.0052	0.0115	0.0147	0.0160	0.0153
	(0.0141)	(0.0141)	(0.0142)	(0.0215)	(0.0215)	(0.0219)	(0.0106)	(0.0105)	(0.0107)
Effect of one treatment SD	-0.0066***	-0.0065***	-0.0063***	0.0005	0.0006	0.0013	0.0028	0.0031	0.0029
	(0.0023)	(0.0023)	(0.0024)	(0.0024)	(0.0024)	(0.0025)	(0.0020)	(0.0020)	(0.0020)
Treatment SD (dS/m)	0.1656	0.1656	0.1656	0.1134	0.1134	0.1134	0.1908	0.1908	0.1908
Δ_s GDD at t		-0.0000	-0.0000		0.0000	-0.0000		0.0000**	0.0000**
		(0.0000)	(0.0000)		(0.0000)	(0.0000)		(0.0000)	(0.0000)
Δ_s KDD at t		-0.0001	-0.0001		-0.0010	-0.0009		-0.0004***	-0.0004***
		(0.0002)	(0.0002)		(0.0037)	(0.0038)		(0.0001)	(0.0001)
Δ_s Soil moisture at t		0.0733	0.0576		0.1884	0.1429		-0.0000	0.0052
		(0.0981)	(0.0987)		(0.1648)	(0.1645)		(0.1063)	(0.1076)
Δ_s Elevation			0.0000			-0.0001			-0.0000
			(0.0000)			(0.0001)			(0.0000)
Δ_s Slope			-0.0028			0.0057			0.0046
			(0.0031)			(0.0048)			(0.0032)
Δ_s Clay			0.0017**			0.0047***			-0.0006
			(0.0008)			(0.0016)			(0.0009)
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.00016	0.00019	0.00022	0.00014	0.00018	0.00034	0.00012	0.00026	0.00028
Observations	98,576	98,576	98,576	55,953	55,953	55,953	81,488	81,488	81,488

Notes: An observation is one spatial municipality pair, crop and baseline campaign t. Baseline campaigns run from 1988 to 2017, and outcomes are measured from t to t+1. Both municipalities have positive planted area at t and positive production at t-1. The dependent variable is the east-minus-west difference in $\text{asinh}(A_{t+1}) - \text{asinh}(A_t)$. Planted areas equal to zero at t+1 are retained. Raw treatment coefficients are per 1 dS/m. The standardized row multiplies the coefficient and standard error by the crop-specific sample treatment SD. Spec. 1 includes year fixed effects; preferred Spec. 2 adds the displayed GDD, KDD and soil-moisture controls; Spec. 3 additionally adds elevation, slope and clay. All specifications use a crop-specific common complete-case sample. Standard errors are clustered by spatial pair. No pair or state fixed effects are included.

Table 30 (continued)

Panel B: Beans, Soybeans and Sugarcane

Crop	Beans			Soybeans			Sugarcane		
Specification	Spec. 1	Spec. 2	Spec. 3	Spec. 1	Spec. 2	Spec. 3	Spec. 1	Spec. 2	Spec. 3
Δ_s Excess salinity at t (dS/m)	-0.0228*	-0.0222*	-0.0176	0.4284	0.4648	0.4128	-0.0071	-0.0077	0.0047
	(0.0123)	(0.0123)	(0.0123)	(0.6441)	(0.6532)	(0.6447)	(0.0265)	(0.0265)	(0.0280)
Effect of one treatment SD	-0.0049*	-0.0047*	-0.0038	0.0065	0.0070	0.0062	-0.0010	-0.0011	0.0007
	(0.0026)	(0.0026)	(0.0026)	(0.0097)	(0.0099)	(0.0097)	(0.0037)	(0.0038)	(0.0040)
Treatment SD (dS/m)	0.2130	0.2130	0.2130	0.0151	0.0151	0.0151	0.1413	0.1413	0.1413
Δ_s GDD at t		-0.0000	0.0000		0.0000	0.0001		0.0000	0.0001
		(0.0000)	(0.0000)		(0.0001)	(0.0002)		(0.0000)	(0.0001)
Δ_s KDD at t		-0.0005***	-0.0004**		-0.0011	-0.0013		-0.0003	-0.0000
		(0.0002)	(0.0002)		(0.0009)	(0.0010)		(0.0002)	(0.0003)
Δ_s Soil moisture at t		0.0140	-0.0176		-0.2023	-0.0767		-0.0819	-0.1322
		(0.1690)	(0.1699)		(0.3085)	(0.3227)		(0.1898)	(0.1944)
Δ_s Elevation			0.0001**			0.0002			0.0001*
			(0.0000)			(0.0002)			(0.0001)
Δ_s Slope			-0.0073**			-0.0382***			0.0024
			(0.0031)			(0.0123)			(0.0045)
Δ_s Clay			0.0028**			-0.0027			0.0020
			(0.0012)			(0.0018)			(0.0015)
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.00012	0.00027	0.00039	0.00065	0.00086	0.00118	0.00023	0.00025	0.00034
Observations	87,422	87,422	87,422	28,302	28,302	28,302	49,676	49,676	49,676

Notes: An observation is one spatial municipality pair, crop and baseline campaign t. Baseline campaigns run from 1988 to 2017, and outcomes are measured from t to t+1. Both municipalities have positive planted area at t and positive production at t-1. The dependent variable is the east-minus-west difference in $\text{asinh}(A_{t+1}) - \text{asinh}(A_t)$. Planted areas equal to zero at t+1 are retained. Raw treatment coefficients are per 1 dS/m. The standardized row multiplies the coefficient and standard error by the crop-specific sample treatment SD. Spec. 1 includes year fixed effects; preferred Spec. 2 adds the displayed GDD, KDD and soil-moisture controls; Spec. 3 additionally adds elevation, slope and clay. All specifications use a crop-specific common complete-case sample. Standard errors are clustered by spatial pair. No pair or state fixed effects are included.

G Total Cultivated Area

The following tables report the six salinity dates estimated jointly, the complete set of estimated controls, and the sensitivity of inference to the cutoff used to compute the Conley spatial standard errors. They correspond to the total-cultivated-area analysis and do not distinguish among crops.

Table 31: Joint Distributed-Lag Model for Annual Total-Cropland Change: Salinity Coefficients

Specification	Year FE	Weather	Weather + soil/topography
Model:	(1)	(2)	(3)
Variables			
Δ_s Mean salinity at t (dS/m)	-0.0425***	-0.0420***	-0.0414***
	(0.0152)	(0.0145)	(0.0146)
Δ_s Mean salinity at $t-1$ (dS/m)	-0.0435**	-0.0436**	-0.0431**
	(0.0190)	(0.0188)	(0.0187)
Δ_s Mean salinity at $t-2$ (dS/m)	-0.0105	-0.0105	-0.0101
	(0.0118)	(0.0119)	(0.0119)
Δ_s Mean salinity at $t-3$ (dS/m)	-0.0023	-0.0021	-0.0020
	(0.0135)	(0.0137)	(0.0136)
Δ_s Mean salinity at $t-4$ (dS/m)	-0.0130	-0.0122	-0.0120
	(0.0146)	(0.0146)	(0.0151)
Δ_s Mean salinity at $t-5$ (dS/m)	0.0224*	0.0233*	0.0230*
	(0.0133)	(0.0132)	(0.0127)
Fixed-effects			
Year	Yes	Yes	Yes
Fit statistics			
R^2	0.00108	0.00198	0.00577
Observations	90,506	90,506	90,506

Conley (200km) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: An observation is one strictly contiguous east-west municipality pair and one baseline year. The dependent variable is the east-minus-west difference in the annual percentage-point change in total physical cropland from t to t+1. Mean salinity at t through t-5 enters jointly in every column. Spec. 2 adds GDD, KDD and soil moisture in t and is preferred. Spec. 3 adds elevation, slope and clay. All columns use the same complete-data sample and include year fixed effects. No pair fixed effects are included. Standard errors are Conley spatial with a 200 km cutoff. All estimated controls are reported in the companion controls table.

Table 32: Joint Distributed-Lag Model for Annual Total-Cropland Change: Estimated Controls

Specification	Year FE	Weather	Weather + soil/topography
Model:	(1)	(2)	(3)
Variables			
Δ_s GDD at t		-3.85×10^{-5}	-2.31×10^{-5}
		(2.42×10^{-5})	(2.83×10^{-5})
Δ_s KDD at t		-0.0005	-0.0003
		(0.0003)	(0.0003)
Δ_s Soil moisture at t		0.1976	0.2104
		(0.2195)	(0.2230)
Δ_s Elevation			0.0002**
			(0.0001)
Δ_s Slope			-0.0576***
			(0.0140)
Δ_s Clay			0.0025
			(0.0039)
Fixed-effects			
Year	Yes	Yes	Yes
Fit statistics			
R^2	0.00108	0.00198	0.00577
Observations	90,506	90,506	90,506

Conley (200km) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: This table reports every estimated control from the same three models and common sample as the salinity-coefficient table. Spec. 2 includes GDD, KDD and soil moisture in t and is preferred. Spec. 3 additionally includes elevation, slope and clay. All columns include year fixed effects and omit pair fixed effects. Standard errors are Conley spatial with a 200 km cutoff.

Table 33: Spatial-Inference Robustness for Annual Cropland Change

Inference	Coefficient	Std. error	P-value
50 km	-0.0649***	(0.0188)	0.001
100 km	-0.0649***	(0.0210)	0.002
200 km	-0.0649***	(0.0161)	0.000
250 km	-0.0649***	(0.0192)	0.001
300 km	-0.0649***	(0.0211)	0.002

Notes: All columns use the Spec. 2 point estimate and sample; only the variance estimator changes. The main cutoff is 200 km. Strict Conley estimates use spherical distance and no positive-semidefinite covariance correction. NA denotes an invalid strict covariance estimate.

G.1 Aquaculture

Aquaculture is kept as a complementary land-use outcome rather than as part of the crop analysis. Table 34 reports the corresponding SFD estimates.

Table 34: Salinity and Aquaculture Area

Outcome: log(Aquaculture area + 1)	Spec 1 (1)	Spec 2 (2)	Spec 3 (3)
Δ_s Mean Salinity ($t-1$, dS/m)	0.1074**	0.1051**	0.1046**
	(0.0464)	(0.0482)	(0.0514)
Δ_s GDD ($t-1$, degree-days)		0.0001	0.0001
		(0.0001)	(0.0002)
Δ_s KDD ($t-1$, degree-days)		0.0011	0.0008
		(0.0009)	(0.0007)
Δ_s Soil Moisture ($t-1$, m^3/m^3)		0.3217	0.3029
		(0.2556)	(0.2222)
Δ_s Slope Gradient (%)			-0.0215
			(0.0135)
Δ_s Elevation (m)			-0.0001
			(0.0002)
Δ_s Clay Content (%)			0.0050
			(0.0058)
R^2	0.00219	0.00759	0.00881
Observations	114,653	114,653	114,653
Year FE	Yes	Yes	Yes

Notes: The dependent variable is the spatial difference in log(aquaculture area + 1). Salinity and agroclimatic controls are measured at $t-1$. Specifications progressively add year fixed effects, agroclimatic controls, and soil and topographic controls. Conley standard errors with a 200 km cutoff are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

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